

NIFTY 100 FINANCIAL INTELLIGENCE PLATFORM

Project Execution Plan · Technical Specification · Feature Catalogue

Project Title:	Nifty 100 Financial Intelligence Platform
Document Type:	Full Technical Specification & Project Execution Plan
Duration:	45 Calendar Days 6 Weekly Sprints 3 Buffer Days
Dataset Universe:	92 Nifty 100 Companies 7 Core + 5 Supplementary Datasets
Total Records:	~11,000+ Data Points Across All Financial Statements
Platform:	12 Modules 120+ Features 50+ Financial KPIs
Document Pages:	50 Pages

1. Document Control & Classification

Administrative Record

Field	Detail
Document Title	Nifty 100 Financial Intelligence Platform — Project Execution Plan
Document ID	DAD-PROJ-001 Version 1.0
Classification	Confidential — Internal Use Only
Status	Approved for Distribution
Project Duration	45 Calendar Days 6 Weekly Sprints 3 Buffer Days
Effective Date	June 2026
Owner	Data Analytics Division — Project Lead
Distribution	Data Analyst Team Members (individually assigned)
Total Pages	50
Datasets	7 Core Datasets + 5 Supplementary Datasets
Companies	92 (Nifty 100 Index Constituents)
KPIs Defined	50+ Financial Key Performance Indicators
Currency Standard	All monetary values in Indian Rupees — Crore (₹ Cr) unless stated

Version History

Version	Date	Change Description
1.0	June 2026	Initial release — full specification for 45-day project
1.1	Post-Sprint 1	Updates based on data quality findings from ETL validation
1.2	Post-Sprint 3	Screeener and scoring refinements after testing
2.0	Day 45	Final delivery version — all acceptance criteria verified

How to Use This Document

Section	Pages	Primary Reader	Purpose
1 — Document Control	2	All	Classification, version history, reader guide
2 — Table of Contents	3	All	Navigation index
3 — Executive Summary	4–5	All	Platform overview, key metrics, outcomes
4 — Project Overview	6–7	All	Objectives, scope, stakeholder RACI
5 — Dataset Catalogue	8–14	Data Engineers	Full data dictionary for all 12 datasets
6 — Data Relationships	15	All	Entity relationships, join patterns, data gaps
7 — System Architecture	16	All	Folder structure, layers, data flow
8–19 — Module Specifications	17–34	Analysts	Feature-by-feature breakdown of all 12 modules
20 — Sprint Plan & Timeline	35–40	All	Day-by-day tasks, exit criteria for each sprint
21 — KPI Reference	41–43	Analytics	Formulas, edge cases, benchmarks for 50+ KPIs
22 — Data Quality Rules	44	Data Engineers	14 validation rules — severity, action, test
23 — API Specification	45	Backend	Optional FastAPI endpoint definitions
24 — Risk Register	46	Project Lead	Risks, likelihood, impact, mitigations
25 — Deliverables Checklist	47	All	What to submit, format, due date, evaluation

Section	Pages	Primary Reader	Purpose
26 — Tech Stack & Standards	48	All	Libraries, versions, code quality gates
27 — Acceptance Criteria	49	All	20 mandatory gates — definition of done
28 — Glossary	50	All	Definitions of all technical terms

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3. Executive Summary

Strategic Overview & Key Metrics

Attribute	Detail
Platform Name	Nifty 100 Financial Intelligence Platform
Core Purpose	Transform raw financial statement data into structured analytics intelligence for 92 Nifty 100 companies
Data Foundation	7 core + 5 supplementary datasets covering P&L, Balance Sheet, Cash Flow, Sectors, Market Cap, Pricing
Analytical Scope	10–13 years of annual financial history per company; 50+ computed KPIs; 12 analytics modules
Primary Outputs	Investment screener, financial health scores, sector benchmarks, automated PDF/Excel reports, Streamlit dashboard
Target Users	Data Analysts — individually assigned to specific modules per this project document
Build Timeline	45 calendar days 6 weekly sprints 3 buffer days
Quality Standard	Production-grade: ≥80% test coverage, full documentation, zero critical bugs at handoff
Currency Standard	All monetary values in Indian Rupees — Crore (₹ Cr) unless explicitly stated otherwise
Index Universe	Nifty 100 — 92 companies after data availability filter applied

Category	Metric	Value	Category	Metric	Value
Data	Core Datasets	7	Platform	Analytics Modules	12
Data	Supplementary Datasets	5	Platform	Total Features	120+
Data	Total Companies	92	Platform	Financial KPIs	50+
Data	P&L Records	1,276	Platform	Screener Filters	18
Data	Balance Sheet Records	1,312	Platform	Pre-built Screens	6
Data	Cash Flow Records	1,187	Platform	Peer Groups	11
Data	Annual Report Links	1,585	Platform	Sector Categories	11
Data	Stock Price Records	5,520	Quality	Test Coverage Target	≥80%
Data	Computed Ratio Records	1,184	Quality	DQ Validation Rules	14
Time	Total Duration	45 Days	Quality	Acceptance Criteria	20

#	Key Outcome	Business Value	Owner Module
01	Unified financial data warehouse for 92 companies	Single source of truth; eliminates manual data gathering	ETL Pipeline
02	50+ computed KPIs per company per year	Instant quantitative assessment without manual calculation	Ratio Engine
03	Multi-parameter investment screener (18 filters)	Reduces analyst time-to-insight from hours to seconds	Screener
04	Composite Financial Health Score (0–100)	Standardised risk triage across the full universe	Health Score
05	Sector benchmarks across 11 broad sectors	Enables sector rotation analysis and relative positioning	Sector Analytics
06	Peer comparison engine for 11 peer groups	Identifies best-in-class within comparable peer companies	Peer Comparison
07	Automated PDF and Excel report generation	Reproducible, consistent output; eliminates manual reporting	Reporting
08	Streamlit dashboard — analyst-facing interface	Zero-code interface for non-programmer stakeholders	Dashboard
09	Annual report repository — 1,585 links	Direct access to primary source documents	Document Repo
10	Rule-based alert system and company watchlist	Proactive KPI deterioration monitoring	Alerts

Sprint	Days	Focus Area	Primary Deliverable	Exit Gate
Sprint 1	1–7	Foundation & ETL	SQLite DB loaded; DQ report	0 critical DQ violations; ≥80% ETL coverage
Sprint 2	8–14	Ratio Engine	50+ KPIs in financial_ratios table	Values validated vs 5 companies ±2%
Sprint 3	15–21	Screener, Score & Sector	Screener, FHS, Sector module	All 3 modules correct; scores 0–100 for all 92

Sprint	Days	Focus Area	Primary Deliverable	Exit Gate
Sprint 4	22–28	Peer / Trend / CF / Docs	4 analytics modules complete	All modules integrated; peer comparison works
Sprint 5	29–35	Dashboard & Reporting	Streamlit app; PDF/Excel generators	App runs; 10+ company PDFs generated
Sprint 6	36–45	Testing, Docs & Delivery	v1.0 tagged release	≥80% coverage; 0 critical bugs; README verified

4. Project Overview & Objectives

Mission, Scope, Stakeholders & Governance

4.1 Mission & Objectives Matrix

Attribute	Statement			
Mission	Build a self-contained, production-grade financial intelligence system enabling analysts to query, screen, score, and compare all 92 Nifty 100 companies.			
Vision	Become the definitive internal reference for Nifty 100 fundamental analysis — reproducible, auditable, and version-controlled from publication.			
Principle	Every insight must be traceable to a formula, every formula to a source column, every column to a dataset file. No black-box numbers.			

ID	Objective	Success Metric	Priority	Sprint
O-01	Data Engineering	All 12 datasets loaded; 0 critical DQ violations	Critical	1
O-02	KPI Computation	50+ KPIs for all company-year combinations	Critical	2
O-03	Investment Screener	18 filter types; 6 pre-built templates return results	Critical	3
O-04	Financial Health Scoring	0–100 score for all 92 companies; correct trends	Critical	3
O-05	Sector Analytics	Median KPIs for all 11 sectors; sector ranking	High	3
O-06	Peer Comparison	Percentile ranks for all 11 peer groups	High	4
O-07	Trend & Growth Analytics	1Y/3Y/5Y/10Y CAGR for revenue, profit, EPS	High	4
O-08	Cash Flow Intelligence	Capital allocation matrix for all 92 companies	High	4
O-09	Dashboard	Streamlit app — 5 pages functional; no bugs	High	5
O-10	PDF Report Generation	Company + sector PDFs for all 92 + 11 sectors	Medium	5
O-11	Excel Exports	Screener export, full universe, comparison exports	Medium	5
O-12	Alerts & Watchlist	Threshold alerts; YoY deterioration detection	Medium	6
O-13	Test Coverage	≥80% unit test coverage across src/	Critical	6
O-14	Documentation	README: fresh setup in <30 min; docs on all functions	High	6

4.2 Scope Boundaries

#	In Scope	#	Out of Scope
S01	Fundamental analysis — P&L, Balance Sheet, Cash Flow	X01	Real-time price feeds or live market data
S02	Historical trend analysis — 10–13 years per company	X02	Technical analysis (RSI, MACD, Bollinger Bands)
S03	50+ financial KPI computation from raw data	X03	Machine learning price prediction models
S04	Investment screener with 18 configurable filters	X04	Options, derivatives, or futures analytics
S05	Financial Health Score (0–100) composite model	X05	User authentication or multi-tenancy
S06	Sector benchmarking across 11 sectors	X06	Cloud deployment or CI/CD infrastructure
S07	Peer comparison for 11 defined peer groups	X07	Insider trading or shareholding pattern data
S08	Automated PDF and Excel report generation	X08	ESG scoring or sustainability metrics
S09	Streamlit dashboard — analyst interface	X09	International or cross-market comparison
S10	Annual report repository — 1,585 links	X10	Portfolio construction or optimisation models
S11	Simulated stock price & market cap datasets	X11	Earnings call transcripts or news sentiment
S12	Rule-based alert system and watchlist	X12	Regulatory filing alerts or SEBI announcements

4.3 Stakeholder Matrix & RACI

Stakeholder	Role	Module Ownership	Engagement
Data Analyst — ETL	Data Engineer	Module 1 (ETL Pipeline)	Daily — Sprints 1–2
Data Analyst — Analytics	KPI Developer	Modules 2, 5, 6, 7, 8, 9	Daily — Sprints 2–4
Data Analyst — Frontend	Dashboard Builder	Module 3, 4 UI (Screener page)	Daily — Sprint 5
Data Analyst — Reporting	Report Developer	Module 10 (PDF/Excel)	Daily — Sprint 5
Data Analyst — QA	Quality Assurance	Test suite, documentation	Daily — Sprint 6
Project Lead	Coordinator	All modules — review & integration	Weekly review meetings

Module	ETL Analyst	Analytics Analyst	Frontend Analyst	Reporting Analyst	QA Analyst
ETL Pipeline	R/A	C	I	I	C
Financial Ratio Engine	C	R/A	I	I	C
Investment Screener	I	R/A	C	I	C
Health Scoring	I	R/A	C	I	C
Sector Analytics	I	R/A	C	I	C
Peer Comparison	I	R/A	C	I	C
Trend & Growth	I	R/A	C	I	C
Cash Flow Intelligence	I	R/A	C	I	C
Dashboard	I	C	R/A	C	C
PDF / Excel Reports	I	C	C	R/A	C
Alerts & Watchlist	I	R	C	I	A
Full Test Suite	C	C	C	C	R/A
Documentation	C	C	C	C	R

R = Responsible | A = Accountable | C = Consulted | I = Informed

5. Dataset Catalogue — Core Datasets

Schema, Data Dictionary, Quality Notes & Usage

Dataset	Sheet	Records	Cols	Time Coverage	Primary Key	Currency
companies.xlsx	Companies	92	12	Current snapshot	id (NSE ticker)	N/A
profitandloss.xlsx	Profit & Loss	1,276	15	FY 2010–2024	(company_id, year)	■ Crore
balancesheet.xlsx	Balance Sheet	1,312	13	FY 2010–2024	(company_id, year)	■ Crore
cashflow.xlsx	Cash Flow	1,187	7	FY 2010–2024	(company_id, year)	■ Crore
analysis.xlsx	Analysis	20	6	Multi-period text	company_id	% strings
documents.xlsx	Documents	1,585	4	Calendar 2010–2024	(company_id, Year)	N/A — URLs
prosandcons.xlsx	Pros & Cons	16	4	Current	id (auto)	N/A — text

Load note: Core files use `pd.read_excel(path, header=1)` — Row 0 is metadata; Row 1 is actual headers. Supplementary files use `header=0`.
Normalise company_id to uppercase stripped NSE ticker on every load.

5.1 companies.xlsx — Master Company Reference

Column	Type	Nullable	Sample Value	Description & Notes
id	VARCHAR	No	TCS	NSE ticker — Primary Key for entire platform. Strip whitespace; uppercase.
company_logo	TEXT	Yes	https://.../TCS.png	URL to company logo image. Some may return 404 — handle gracefully.
company_name	VARCHAR	No	Tata Consultancy Services Ltd	Full legal name. Some entries contain embedded \n — strip on load.
chart_link	TEXT	Yes	https://in.tradingview.com/...	TradingView chart URL. For display only; not used in computation.
about_company	TEXT	Yes	TCS is a leading IT...	Business description. Length: 50–600 chars. Display in company card.
website	TEXT	Yes	https://www.tcs.com/	Official corporate website URL.
nse_profile	TEXT	Yes	https://nseindia.com/...	NSE India equity profile page URL.
bse_profile	TEXT	Yes	https://bseindia.com/...	BSE India stock page URL.
face_value	NUMERIC	No	1	Share face value in ■. Common: 1, 2, 5, 10. Used in book value calculation.
book_value	NUMERIC	Yes	157.40	Book value per share in ■ (latest). Display only; compute from BS for analytics.
roce_percentage	NUMERIC	Yes	64.30	ROCE % (pre-computed). Use for display; Ratio Engine values for analytics.
roe_percentage	NUMERIC	Yes	0.52	ROE % (pre-computed). Note: TCS shows 0.52 — anomaly; use Ratio Engine.

5.2 profitandloss.xlsx — Annual Profit & Loss Statements

Column	Type	Nullable	Sample (TCS FY23)	Formula / Notes
id	INTEGER	No	61	Row ID — not analytically meaningful. Use composite key.
company_id	VARCHAR	No	TCS	FK → companies.id. Normalise to uppercase stripped ticker.
year	VARCHAR	No	Mar-23	Financial year label. Standardise to 'YYYY-MM' on load via <code>normalize_year()</code> .
sales	NUMERIC	No	225,458	Net revenue / total income. Must be > 0. ■ Crore.
expenses	NUMERIC	No	176,924	Total operating expenses. ■ Crore.
operating_profit	NUMERIC	No	48,534	EBITDA. Cross-validate: must equal sales – expenses within ±1%.
opm_percentage	NUMERIC	No	21.5	OPM %. Validate: $= \text{operating_profit} / \text{sales} \times 100 \pm 1\%$. Flag if diff >1%.
other_income	NUMERIC	Yes	3,800	Non-operating income (dividends received, interest). ■ Crore.
interest	NUMERIC	Yes	0	Finance costs. 0 for debt-free companies. ■ Crore.
depreciation	NUMERIC	Yes	5,800	D&A. ■ Crore. EBIT = operating_profit – depreciation.

Column	Type	Nullable	Sample (TCS FY23)	Formula / Notes
profit_before_tax	NUMERIC	Yes	46,534	PBT = EBITDA + other_income – interest – D&A. ■ Crore.
tax_percentage	NUMERIC	Yes	25.0	Effective tax rate %. Verify: (PBT – PAT) / PBT × 100.
net_profit	NUMERIC	Yes	34,990	PAT. Can be negative. ■ Crore.
eps	NUMERIC	Yes	95.30	Earnings Per Share in ■. Must be > 0 if net_profit > 0.
dividend_payout	NUMERIC	Yes	45.0	Dividend payout ratio %. Can exceed 100% in low-profit years.

5.3 balancesheet.xlsx — Annual Balance Sheet

Column	Type	Nullable	Sample (ABB 2024)	Formula / Notes
id	INTEGER	No	136	Row ID.
company_id	VARCHAR	No	ABB	FK → companies.id.
year	VARCHAR	No	Mar-24	Standardise to YYYY-MM.
equity_capital	NUMERIC	No	21	Paid-up share capital. Small for large caps. ■ Crore.
reserves	NUMERIC	Yes	626	Reserves & surplus. Total Equity = equity_capital + reserves.
borrowings	NUMERIC	Yes	0	Total debt (short + long term). 0 = debt-free. ■ Crore.
other_liabilities	NUMERIC	Yes	260	Trade payables + other current liabilities. Proxy for CL in ROCE. ■ Crore.
total_liabilities	NUMERIC	No	907	Sum of equity + all liabilities. Must equal total_assets ±1%.
fixed_assets	NUMERIC	Yes	109	Net fixed assets (after D&A). ■ Crore.
cwip	NUMERIC	Yes	1	Capital Work In Progress. Add to fixed_assets for gross block proxy.
investments	NUMERIC	Yes	0	Long-term investments and securities. Used as liquid proxy for net debt.
other_asset	NUMERIC	Yes	798	Current assets + other assets. Catch-all non-fixed.
total_assets	NUMERIC	No	907	Sum of all asset-side items. Must equal total_liabilities ±tolerance.

5.4 cashflow.xlsx — Annual Cash Flow Statements

Column	Type	Sign Convention	Healthy Signal	Notes
id	INTEGER	—	—	Row ID.
company_id	VARCHAR	—	—	FK → companies.id.
year	VARCHAR	—	—	Standardise to YYYY-MM.
operating_activity	NUMERIC	Positive = good	CFO > 0 consistently	Cash generated from core operations. Primary health indicator. ■ Crore.
investing_activity	NUMERIC	Negative = normal	Negative = investing in growth	Cash used for CapEx, acquisitions, investments. ■ Crore.
financing_activity	NUMERIC	Variable	Negative = debt repayment / Dividends	Debt raise/repay, equity issuance, dividends paid. ■ Crore.
net_cash_flow	NUMERIC	Pos = cash grew	Positive preferred	= CFO + CFI + CFF. Validate: matches sum ±10 Crore. ■ Crore.

5.5 analysis.xlsx — Pre-Computed Growth Metrics (Partial Coverage)

Column	Type	Sample Value	Parsing Logic	Usage
id	INTEGER	1	Direct int cast	Row ID.
company_id	VARCHAR	HDFCBANK	Normalise ticker	FK → companies.id.
compounded_sales_growth	TEXT	10 Years: 21%	Regex: r'(\d+)\s*Years?:?(?!\d.]+)%'	Display + cross-check after parsing
compounded_profit_growth	TEXT	5 Years: 6%	Same pattern	Display + cross-check
stock_price_cagr	TEXT	10 Years: 15%	Same pattern	Display only — compute from market_cap dataset
roe	TEXT	10 Years: 17%	Same pattern	Display only — use Ratio Engine ROE for analytics

Coverage gap: Only ~8 companies have entries. Compute growth metrics for all 92 via the Ratio Engine (Sprint 2). Use analysis.xlsx values only for display cross-check.

5.6 documents.xlsx — Annual Report Repository

Column	Type	Nullable	Sample Value	Notes
id	INTEGER	No	1	Row ID.
company_id	VARCHAR	No	ABB	FK → companies.id.

Column	Type	Nullable	Sample Value	Notes
Year	INTEGER	No	2024	Calendar year of annual report. Note: capital 'Y'. Cast to int.
Annual_Report	TEXT	Yes	https://bseindia.com/...pdf	Direct PDF URL on BSE India. Validate with requests.head(). Some return 4

5.7 prosandcons.xlsx — Qualitative Investment Insights

Column	Type	Nullable	Sample Value	Notes
id	INTEGER	No	1	Row ID.
company_id	VARCHAR	No	HDFCBANK	FK → companies.id.
pros	TEXT	Yes	Company is expected to give good positive observation.	One per row. Multiple rows per company allowed.
cons	TEXT	Yes	Stock trading at 2.76× book value	Risk observation. One per row.

Coverage gap: Only 16 records covering ~8 companies. Sprint 3 task: auto-generate pros/cons for all 92 companies using KPI threshold rules.
Examples: ROE>20% sustained → pro; D/E>2.0 → con; FCF negative 3yr → con.

6. Supplementary Datasets

Created to Augment
Core Analysis —
Location: supporting
datasets/

File	Rows	Created From	Analytical Purpose
sectors.xlsx	92	Manual GICS-style mapping of all 92 companies	Sector/sub-sector aggregation, relative scoring, sector filter in screener
stock_prices.xlsx	5,520	Simulated monthly OHLCV — Jan 2020 to Dec 2024	Price trend and charts, monthly return calculation, FCF yield denominator
market_cap.xlsx	552	Simulated annual P/E, P/B, EV/EBITDA, Market Cap	Valuation screening (P/E, P/B), EV/EBITDA comp, dividend yield d
financial_ratios.xlsx	1,184	Computed from P&L + Balance Sheet + Cash Flow	Primary computed KPI table; screener backbone; health score inputs
peer_groups.xlsx	56	Manually defined 11 peer groups + benchmark	Peer comparison engine; relative percentile ranking

6.1 sectors.xlsx — Company Sector Mapping

Column	Values / Range	Notes
company_id	NSE Ticker (92 unique)	FK → companies.id. One row per company.
broad_sector	11 macro sectors	Financials, Energy, IT, Consumer Discretionary, Consumer Staples, Healthcare, Materials, Industrials, Communication Services, Real Estate, Conglomerates / Other
sub_sector	33 sub-sectors	e.g. Private Banks, IT Services, Pharmaceuticals, Steel, FMCG, etc.
index_weight_pct	0.2 – 4.5%	Estimated Nifty 100 index weight. For display only; not used in KPI calculations.
market_cap_category	Large Cap / Mid Cap	Classification at dataset creation time.

Broad Sector	Sub-Sectors Included	Companies
Financials	Private Banks, Public Sector Banks, Life Insurance, General Insurance, Consumer Finance	10
Energy	Oil & Gas Refining, Oil & Gas Exploration, Power & Utilities, Renewable Energy, Gas Distribution, Power Transmission	15
Information Technology	IT Services, Internet & Platforms	6
Consumer Discretionary	Automobiles, Two Wheelers, Auto Ancillaries, Retail, Gems & Jewellery, Airlines, Travel & Tourism	12
Consumer Staples	FMCG, Food Products, Food & Beverages, Personal Products	8
Healthcare	Pharmaceuticals	5
Materials	Steel, Metals & Mining, Cement, Specialty Chemicals, Paints & Coatings	8
Industrials	Capital Goods, Defence Electronics, Defence & Aerospace, Engineering & Construction, Consumer Electricals	9
Communication Services	Telecommunications, Internet & Platforms	2
Real Estate	Real Estate, Infrastructure	3
Conglomerates / Other	Diversified Conglomerates, Holding Companies	5

6.2 stock_prices.xlsx — Monthly OHLCV Price History (Simulated)

Column	Type	Range / Format	Notes
company_id	VARCHAR	NSE Ticker	FK → companies.id.
date	VARCHAR	YYYY-MM-DD	First calendar day of each month. Covers Jan 2020 – Dec 2024 (60 months).
open_price	NUMERIC	■5 – ■18,000	Simulated monthly opening price. For price trend and return charts.
high_price	NUMERIC	■5 – ■19,000	Monthly high. Always ≥ open and ≥ close.
low_price	NUMERIC	■4 – ■17,500	Monthly low. Always ≤ open and ≤ close.
close_price	NUMERIC	■5 – ■18,000	Monthly closing price. Primary field for return calculations.
volume	INTEGER	100K – 50M	Simulated monthly traded volume in shares.
adjusted_close	NUMERIC	■5 – ■18,000	Same as close_price (no dividend adjustment in simulation).

6.3 market_cap.xlsx — Annual Valuation Multiples

Column	Type	Range	Notes
company_id	VARCHAR	NSE Ticker	FK → companies.id.
year	INTEGER	2019–2024	Calendar year. One row per company per year.
market_cap_core	NUMERIC	■5K–■15L Cr	Market capitalisation. Simulated. Use for EV, P/E, FCF yield denominator.
enterprise_value_core	NUMERIC	■4K–■17L Cr	EV = Mkt Cap + Net Debt (proxy). Simulated.
pe_ratio	NUMERIC	8–80×	Price-to-Earnings. Simulated around sector norms.
pb_ratio	NUMERIC	0.5–15×	Price-to-Book. Cross-check vs book_value in companies.xlsx.
ev_ebitda	NUMERIC	5–40×	EV / EBITDA multiple. Use for sector valuation comparison.
dividend_yield_pct	NUMERIC	0–4.5%	Annual dividend yield. Simulated.

6.4 financial_ratios.xlsx — Pre-Computed KPI Table

Column	Source	Formula / Notes
company_id, year	All	Composite PK.
net_profit_margin_pct	P&L	net_profit / sales × 100. Negative allowed.
operating_profit_margin_pct	P&L	From opm_percentage source field. Cross-validate vs computed.
return_on_equity_pct	P&L+BS	net_profit / (equity_cap + reserves) × 100.
debt_to_equity	BS	borrowings / (equity_cap + reserves). 0 = debt-free.
interest_coverage	P&L	(op_profit + other_income) / interest. None if interest = 0.
asset_turnover	P&L+BS	sales / total_assets.
free_cash_flow_cr	CF	operating_activity + investing_activity (■ Cr).
capex_cr	CF	abs(investing_activity) — CapEx proxy (■ Cr).
earnings_per_share	P&L	eps source field (■).
book_value_per_share	BS+Co	(equity + reserves) / (equity_cap / face_value) (■).
dividend_payout_ratio_pct	P&L	dividend_payout source field (%).
total_debt_cr	BS	borrowings (■ Cr).
cash_from_operations_cr	CF	operating_activity (■ Cr).

6.5 peer_groups.xlsx — Peer Comparison Groups

Peer Group	Members	Benchmark
Private Banks	HDFCBANK, ICICIBANK, AXISBANK, KOTAKBANK, INDUSINDBK	HDFCBANK
Public Banks	SBIN, BANKBARODA, CANBK, PNB	SBIN
IT Services	TCS, INFY, HCLTECH, TECHM, LTIM	TCS
Pharmaceuticals	SUNPHARMA, CIPLA, DRREDDY, DIVISLAB, TORNTPHARM	SUNPHARMA
Automobiles	MARUTI, TATAMOTORS, M&M, BAJAJ-AUTO, EICHERMOT, HEROMOTOCO, TVS	MARUTI
Life Insurance	LICI, HDFCLIFE, SBILIFE, ICICIPRULI	LICI
Oil & Gas	RELIANCE, ONGC, BPCL, IOC, GAIL	RELIANCE
Power & Utilities	NTPC, POWERGRID, TATAPOWER, ADANIPOWER, NHPC, JSWENERGY, ADANI	GREEN
Steel & Metals	TATASTEEL, JSWSTEEL, JINDALSTEL, HINDALCO	TATASTEEL
FMCG	HINDUNILVR, ITC, BRITANNIA, DABUR, NESTLEIND, GODREJCP, TATACONSUM	HINDUNILVR

Peer Group	Members	Benchmark
Consumer Finance	BAJFINANCE, CHOLAFIN, SHRIRAMFIN	BAJFINANCE

7. Data Relationships & Entity Map

Foreign Keys, Join Paths & Coverage Matrix

7.1 Entity-Relationship Summary

Parent Table	Parent Key	Child Table	Child Key	Cardinality	Join Note
companies	id (NSE ticker)	profitandloss	company_id	1 : N	Up to 15 annual rows per company
companies	id	balancesheet	company_id	1 : N	Up to 15 annual rows per company
companies	id	cashflow	company_id	1 : N	Up to 14 annual rows per company
companies	id	analysis	company_id	1 : 1	Partial — ~8 companies have entries
companies	id	documents	company_id	1 : N	1–20 annual report links per company
companies	id	prosandcons	company_id	1 : N	Partial — ~8 companies with entries
companies	id	sectors	company_id	1 : 1	All 92 companies covered
companies	id	stock_prices	company_id	1 : N	60 monthly rows per company
companies	id	market_cap	company_id	1 : N	6 annual rows (2019–2024)
companies	id	financial_ratios	company_id	1 : N	Up to 14 annual rows per company
companies	id	peer_groups	company_id	M : N	A company can be in multiple groups

7.2 Dataset Coverage Matrix

Dataset	Companies Covered	Years Available	Completeness	Data Source
companies	92 / 92 (100%)	Snapshot only	Complete	Screener.in scrape
profitandloss	92 / 92	FY 2010–2024	~95%	Screener.in / BSE filings
balancesheet	92 / 92	FY 2010–2024	~97%	Screener.in / BSE filings
cashflow	92 / 92	FY 2010–2024	~91%	Screener.in / BSE filings
analysis	~8 / 92 (9%)	Multi-period	Partial	Screener.in pre-computed metrics
documents	~75 / 92 (82%)	Calendar 2010–24	Partial	BSE India Annual Reports portal
prosandcons	~8 / 92 (9%)	Snapshot only	Partial	Screener.in qualitative section
sectors	92 / 92	Snapshot	Complete	Created — GICS-style manual mapping
stock_prices	92 / 92	Jan 2020–Dec 2024	Complete	Created — simulated OHLCV
market_cap	92 / 92	2019–2024	Complete	Created — simulated valuation
financial_ratios	92 / 92	FY 2010–2024	~93%	Created — computed from raw files
peer_groups	46 / 92 (50%)	Snapshot	Partial	Created — manually defined groups

7.3 Standard Join Pattern — Python / SQL

Scenario	Join Logic
P&L + Balance Sheet for one year	SELECT p.*, b.borrowings, b.total_assets FROM profitandloss p JOIN balancesheet b USING (company_id, year)
Full profile with sector	SELECT c.id, c.company_name, s.broad_sector, s.sub_sector FROM companies c JOIN sectors s ON c.id = s.company_id
CFO + Net Profit health check	SELECT p.company_id, p.year, p.net_profit, f.operating_activity FROM profitandloss p JOIN cashflow f USING (company_id, year)
Valuation multiples with earnings	SELECT m.*, p.net_profit, p.eps FROM market_cap m JOIN profitandloss p USING (company_id, year)
Peer group ranking	SELECT r.company_id, r.return_on_equity_pct, RANK() OVER (PARTITION BY pg.peer_group_name ORDER BY r.return_on_equity_pct) AS rank FROM return_on_equity r JOIN peer_groups pg ON r.company_id = pg.company_id

8. System Architecture

7-Layer Data Platform
with 12 Analytical
Modules

Layer	Layer Name	Components	Technology
L1	Raw Data Ingestion	Excel loader, URL validator, schema validator	Python — pandas, openpyxl, requests
L2	ETL & Normalisation	Year normaliser, ticker normaliser, dedup engine	Python — pandas, custom ETL modules
L3	Persistent Storage	SQLite database (10 tables) — core + supporting	SQLite 3.x — single-file, zero-config
L4	Analytics Engine	Ratio engine, screener, comparator, capital classifier	Python — pandas, NumPy, scipy.stats
L5	Intelligence Layer	Peer percentiles, health scorer, NLP summarizer	Python — scikit-learn, regex, NLTK
L6	Reporting Layer	PDF reports, Excel exports, chart generation	ReportLab, openpyxl, matplotlib, Plotly
L7	Dashboard & API	Streamlit app (8 screens), REST API (16 endpoints)	Streamlit, FastAPI, Uvicorn

8.1 Technology Stack

Category	Library / Tool	Version	Purpose
Data	pandas	≥ 2.0	DataFrame operations, Excel read/write, pivot tables
Data	NumPy	≥ 1.24	Vectorised KPI computation, growth rate arrays
Data	openpyxl	≥ 3.1	Read .xlsx files with header=1 for core datasets
Database	SQLite	3.x	Single-file relational store; zero server overhead
Analytics	scipy	≥ 1.11	Statistical distributions, hypothesis tests (t-test, Mann-Whitney)
Analytics	scikit-learn	≥ 1.3	KMeans clustering, percentile ranking, scaler
Visualisation	matplotlib	≥ 3.7	Static charts for PDF reports
Visualisation	Plotly	≥ 5.18	Interactive charts for Streamlit dashboard
Dashboard	Streamlit	≥ 1.30	Multi-page web dashboard — no frontend code needed
API	FastAPI	≥ 0.110	REST API server with OpenAPI docs auto-generated
API	Uvicorn	≥ 0.27	ASGI server to run FastAPI
PDF	ReportLab	≥ 4.1	PDF document generation (all reports)
NLP	NLTK	≥ 3.8	Text tokenisation for qualitative analysis + pros/cons generation
NLP	regex	stdlib	Pattern extraction from analysis.xlsx text fields
Testing	pytest	≥ 7.4	Unit tests for ETL, KPI formulas, edge cases
Dev	Jupyter	≥ 7.0	Exploratory analysis notebook (not in production path)
Dev	black	≥ 24.0	Code formatting — PEP8 compliance
Dev	ruff	≥ 0.4	Linting — style and import checks
Dev	pre-commit	≥ 3.7	Git hooks: black + ruff run before every commit
Env	python-dotenv	≥ 1.0	Config via .env file — paths, thresholds, flags

9. Module Specifications — Modules 1–4

Data Foundation,
Analytics Engine,
Screener & Peer
Comparison

Module 1: Data Ingestion & ETL

Reads all 7 core + 5 supplementary Excel files, normalises fields, validates schemas, removes duplicates, and loads into a 10-table SQLite database. The foundation for all downstream modules.

Sprint: Sprint 1 (Days 1–7)

#	Feature	Acceptance Criterion	Output
1.1	Excel file loader with header=1 support	Loads all 7 core files with correct column headers; no merged cells	DataFrames
1.2	Ticker normaliser	company_id.str.strip().str.upper() on all tables; No integrity issues	Normalised tables
1.3	Year label standardiser	Mar-23 → 2023-03 via normalize_year(); Mar / Dec for year endings	Defined year labels
1.4	Schema validator (16 rules)	Zero schema failures on reference dataset; fail validation_dq.csv	validation_dq.csv
1.5	Deduplication engine	No duplicate (company_id, year) pairs in time-series tables	Clean tables
1.6	SQLite loader	All 10 tables written; rowcounts match source; Fk by DB constraints enabled	10 tables
1.7	Load audit log	load_audit.csv: table, rows_in, rows_out, rejected, duration, runtime_s	Audit log

Output	Detail
nifty100.db (SQLite)	10 tables: companies, profitandloss, balancesheet, cashflow, analysis, documents, prosandcons, sectors, market_cap, stock_prices
load_audit.csv	Per-table load statistics for QA review. Row count, rejection count, runtime.
validation_failures.csv	All DQ rule violations with company_id, year, field, issue, severity (CRITICAL/WARNING).

Module 2: Financial Ratio Engine

Computes 50+ KPIs for every company-year combination across all 14 available years. Handles all edge cases: division by zero, negative equity, debt-free companies, bank/NBFC D/E carve-out, and CAGR sign anomalies.

Sprint: Sprint 2 (Days 8–14)

#	Feature / KPI	Formula	Edge Case
2.1	Net Profit Margin	net_profit / sales × 100	None if sales = 0
2.2	Operating Profit Margin	operating_profit / sales × 100	Cross-check vs opm_percentage
2.3	Return on Equity (ROE)	net_profit / (equity + reserves) × 100	None if equity ≤ 0
2.4	Return on Capital (ROCE)	(EBIT) / (equity + reserves + borrowings) × 100	Exclude NBFC/Banks from D/E flag
2.5	Debt-to-Equity	borrowings / (equity + reserves)	0 for debt-free; flag >5 for non-financials
2.6	Interest Coverage	(op_profit + other_income) / interest	None if interest = 0; 999 displayed as 'Debt Free'
2.7	Free Cash Flow	CFO + CFI	Negative allowed; flag if negative 3yr consecutive
2.8	CAGR (Revenue, PAT, EPS)	((end/start)^(1/n) – 1) × 100	Turnaround flag if base < 0 and end > 0
2.9	Asset Turnover	sales / total_assets	None if total_assets = 0
2.10	Capital Allocation Class	Sign pattern of CFO/CFI/CFF	8 patterns → labels: Reinvest, Return, Distress, etc.

Output	Detail
financial_ratios table (SQLite)	14 computed KPI columns per company-year; ~1,184 rows minimum
capital_allocation.csv	company_id, year, CFO_sign, CFI_sign, CFF_sign, pattern_label
ratio_edge_cases.log	Log of every CAGR turnaround, debt-free substitution, or division-by-zero

Module 3: Company Screener & Filter Engine

Multi-criteria stock screener with 15+ filterable metrics. Produces ranked result sets. Analyst configures thresholds via YAML config; no code changes needed to add new screening presets.

Sprint: Sprint 3 (Days 15–21)

#	Feature	Criteria / Logic	Output
3.1	Preset screeners (6 templates)	Quality: ROE>15%, D/E<1, FCF>0. Value: P/E<20, EV/EBITDA<15, PAT CAGR >20% 5yr. Dividend: Yield>2%	Ranking of companies
3.2	Custom filter builder	Analyst defines thresholds per metric in YAML	Config (screener_config.yaml)
3.3	Ranking engine	RANK() OVER sector partition on composite score (50% profitability, 30% growth, 20% valuation)	Score (0-100)
3.4	Sector-relative screener	Same thresholds normalised within broad_sector flag adjusts to #2σ or bottom decile)	Decile
3.5	Screener export (Excel + CSV)	Top-N results with all KPIs; sortable, formatted	screener_output.xlsx
3.6	Multi-year trend filter	Filter companies where metric improved consecutively for N years	Trend filter

Output	Detail
screener_output.xlsx	Top results for each preset with 20+ KPIs displayed, colour-coded by threshold
screener_config.yaml	All screening thresholds; analysts edit this file to create custom screens
screener API endpoint	GET /api/v1/screener?min_roe=15&max_de=1§or=IT → JSON response

Module 4: Peer Comparison Engine

Compares companies within their peer group (from peer_groups.xlsx) across 20 metrics. Computes within-group percentile rank for every metric. Produces radar charts and comparison tables.

Sprint: Sprint 3 (Days 15–21)

#	Feature	Logic	Output
4.1	Intra-group percentile rank	PERCENT_RANK() OVER (PARTITION BY peer_group ORDER BY metric)	Rank (0-100) per metric
4.2	Multi-metric radar chart	8 axes: ROE, ROCE, NPM, D/E, FCF, PAT CAGR, Revenue CAGR, EPS CAGR	Radar chart
4.3	Side-by-side comparison table	Up to 5 companies: all 14 ratios, 5-year CAGR, Capitalised return pattern	Comparison table
4.4	Benchmark gap analysis	Metric_value vs benchmark_company in peer_group; green=above bench, red=below	Gap analysis
4.5	Best-in-class detection	Company in top quartile (≥75th pctile) for ≥6 of 20 metrics → 'Best in Class' badge	Best-in-class flag
4.6	Weak company detection	Bottom quartile (≤25th pctile) for ≥4 of 10 metrics → 'Flag Watch List' flag	Weak company flag

Output	Detail
peer_comparison.xlsx	One sheet per peer group; 20 metrics, colour-coded percentile cells
radar_charts/	One radar PNG per company showing performance vs peer group average
peer_percentiles table	SQLite: company_id, peer_group, metric, value, percentile_rank, year

10. Module Specifications — Modules 5–8

Dashboard, Valuation,
Cash Flow Intelligence
& Reports

Module 5: Interactive Streamlit Dashboard

8-screen web application. No authentication required for MVP. All screens load from SQLite. Deployed locally via `streamlit run app.py`. All charts are Plotly interactive; all tables are sortable.

Sprint: Sprint 4 (Days 22–28)

#	Screen	Key Widgets / Features	Data Source
5.1	Home / Overview	Nifty 100 summary KPIs: avg ROE, median P/E, corporate credit limit, market breadth banner.	market_cap, nifty100
5.2	Company Profile	Search by name/ticker. Company card: logo, description, key tiles (6 metrics), P&L chart, BS chart, CF bar chart.	companies, financials
5.3	Financial Screener	Sidebar sliders for 10 metrics. Live-updating results table. CSV download button.	financials
5.4	Peer Comparison	Select peer group → radar chart + side-by-side table toggle between years.	companies, financials
5.5	Trend Analysis	Select company + metric → 10-year sparkline + primary balance sheet line. Multi-metric overlay mode.	financials
5.6	Sector Analysis	Sector selector → bubble chart (revenue vs ROE, size = mkt cap), sector market cap bar chart.	sector_data, companies
5.7	Capital Allocation Map	Treemap of 92 companies by capital pattern (8 financial ratios: CF, P/B, P/E, D/E, D/P, D/F, D/N, D/S). Company list.	financials, companies
5.8	Annual Reports	Company → year → clickable BSE PDF link. Badge for missing reports.	companies, reports

Output	Detail
app.py + pages/	Streamlit multi-page app. One file per screen under pages/ directory.
dashboard_guide.md	Analyst guide: how to navigate each screen, filter, and export data.

Module 6: Valuation & Market Data Module

Uses market_cap.xlsx to compute and display valuation metrics. Provides historical P/E, P/B, EV/EBITDA trends. Flags companies trading above historical average multiples (potentially overvalued).

Sprint: Sprint 4 (Days 22–28)

#	Feature	Logic	Output
6.1	P/E Trend Chart	Line chart of P/E ratio across 2019–2024 per company.	Plotly chart
6.2	P/B vs ROE Scatter	X=P/B, Y=ROE, size=Mkt Cap, colour=sector for 92 companies.	Scatter plot PNG
6.3	EV/EBITDA Comp Table	Current EV/EBITDA vs 5yr median, vs sector median. Flagged if >20% above sector.	Table with 3 columns
6.4	Dividend Yield Ranker	Rank 92 companies by dividend yield. Filter by Ranked list threshold.	Ranked list
6.5	FCF Yield Calculator	FCF / Market_Cap × 100. Rank descending. High FCF yield as value signal.	Table with 2 columns
6.6	Overvaluation Flag	P/E > (sector_median × 1.5) → 'Caution' badge. P/B > (sector_median × 0.7) → 'Discount' badge.	Badges on table

Output	Detail
valuation_summary.xlsx	All 92 companies with current + 5yr median multiples, sector rank, badges
valuation_flags.csv	List of 'Caution' and 'Discount' flagged companies with rationale

Module 7: Cash Flow Intelligence Module

Deep analysis of cash flow quality, CapEx intensity, FCF generation, and capital allocation patterns. Identifies companies with strong FCF compounding over 5+ years.

Sprint: Sprint 4–5 (Days 22–35)

#	Feature	Logic	Output
7.1	CFO Quality Score	CFO/PAT ratio > 1.0 across 5yr avg = 'High Quality Bridge'. <0.5 = 'Accrual Risk'.	Quality Bridge
7.2	CapEx Intensity	CapEx / Revenue %. Light (<3%): asset-light. Heavy (>8%) capital intensive.	Category (Light/Heavy)
7.3	FCF CAGR	FCF CAGR over 5yr and 10yr. Same CAGR rule as PAT CAGR (turnaround flag).	FCF CAGR
7.4	FCF Conversion Rate	FCF / EBITDA. >60% = efficient. <30% = CapEx heavy tier	Efficiency tier
7.5	Debt Repayment Detection	CFF < 0 AND borrowings declining YoY → 'Debt Evergreening' flag.	Debt Evergreening flag
7.6	Distress Pattern	CFO < 0 AND CFF > 0 (equity/debt raise to fund operations) → 'Distress Signal'.	Distress Signal
7.7	Capital Allocation Matrix	8 CFO/CFI/CFF sign patterns → descriptive labels (e.g., 'Debt Repaid', 'Shareholder Returns, etc.)	Capital Allocation

Output	Detail
cashflow_intelligence.xlsx	All companies with CFO quality score, FCF CAGR, CapEx intensity, capital allocation label
distress_alerts.csv	Companies flagged for distress pattern with supporting data

Module 8: Automated PDF Report Generator

Generates company-level and portfolio-level PDF reports on demand using ReportLab. Two report types: Company Tearsheet (2 pages per company) and Portfolio Summary (all 92 companies, 1 page each).

Sprint: Sprint 5 (Days 29–35)

#	Report Type	Contents	Frequency
8.1	Company Tearsheet (2-page)	Page 1: KPI tiles, 10yr revenue & profit bar, ROIC/Patent and Div Line Page 2: BS composition stacked bar, CF	On Demand
8.2	Portfolio Summary (1-page/co)	All 92 companies in one PDF: company name, Weekly batch (Sprint 5) arrow (↑ ↓ →) for 3yr direction.	Weekly batch (Sprint 5)
8.3	Sector Report	Sector header page + all companies in sector. Summary table KPI table. Best/worst in sector highlighted.	Monthly batch
8.4	Screener Output Report	Formatted PDF of screener results with charts, Sorted by composite score.	On Demand
8.5	Peer Group Report	Radar charts + side-by-side table for each of 11 Weekly maps.	Weekly maps

Output	Detail
reports/tearsheets/	92 PDFs — one per company. Filename: <TICKER>_tearsheet.pdf
reports/portfolio/	portfolio_summary_YYYYMMDD.pdf — all 92 companies in one document
reports/sector/	11 sector PDFs. Filename: <SECTOR>_report_YYYYMMDD.pdf

11. Module Specifications — Modules 9–12

NLP, Clustering, API & Testing

Module 9: NLP & Qualitative Analysis Module

Parses and enriches qualitative text fields: about_company, pros, cons, analysis strings. Auto-generates pros/cons for companies with missing records using KPI-threshold rules. Extracts CAGR numbers from analysis.xlsx text fields.

Sprint: Sprint 5 (Days 29–35)

#	Feature	Logic	Output
9.1	Analysis text parser	Regex <code>r'(\d+)\s*Years?:?\s*([d.]*)\s*%'</code> on compound sentences to extract period and value. Parsed sales growth, profit growth, CAGR fields.	
9.2	Auto pros/cons generator	Rule engine: 12 pro rules (ROE>20%, FCF positive 5yrs/500 crachos, CAGR>15%, etc.) + 12 con rules (D/E>100%, ROE<10%, etc.).	
9.3	Business description tagger	Classify about_company text into sector tags using keyword matching. Confirm vs sectors.xlsx.	
9.4	Sentiment scorer	Basic polarity scoring on pros (positive) and cons (negative) text using NLTK SentimentIntensityAnalyzer.	
9.5	CAGR cross-validator	Compare parsed analysis.xlsx CAGR vs Ratio Enterprise Income CAGR. Flag >5% divergence.	
Output		Detail	
pros_cons_generated.csv		company_id, type (pro/con), rule_triggered, text, confidence_pct. Auto-fills gaps for 84 companies.	
analysis_parsed.csv		Structured: company_id, metric_type, period_years, value_pct for all analysis.xlsx rows.	

Module 10: Statistical Analysis & Clustering Module

Uses scikit-learn for KMeans clustering of companies by financial profile. Computes portfolio-level statistics, correlation matrices, and outlier detection. Provides insight into hidden company groupings.

Sprint: Sprint 5–6 (Days 29–42)

#	Feature	Logic	Output
10.1	KMeans clustering (5 clusters)	Features: ROE, D/E, Revenue CAGR, FCF CAGR, P/E, Std dev. Company_id, cluster_id (0–4). Cluster method to KMeans.	Cluster Profile Table: High-Quality Growth, Defensive Dividend, etc. Correlation Heatmap output. Flag: Company-level outlier report. Portfolio P10/P25/P50/P75/P90 percentile table.
10.2	Cluster profiling	Mean/median of each feature per cluster. Assign descriptive labels.	
10.3	Correlation matrix	Pearson correlation of 10 KPIs across all 92 companies.	
10.4	Outlier detection	Z-score per metric per sector. Z >3 → 'Outlier' flag.	
10.5	Portfolio statistics	Distribution of ROE, D/E, P/E across Nifty 100.	
Output		Detail	
cluster_labels.csv		company_id, cluster_id (0–4), cluster_name, distance_from_centroid	
portfolio_stats.csv		Metric, P10, P25, P50, P75, P90, Mean, Std for all 92 companies	
outlier_report.csv		company_id, metric, value, z_score, sector, sector_mean, sector_std	

Module 11: REST API Module

FastAPI server exposing 16 endpoints. Auto-generates OpenAPI documentation at /docs. All responses are JSON. API is consumed by the Streamlit dashboard for real-time data queries.

Sprint: Sprint 6 (Days 36–42)

#	Endpoint	Method	Description
11.1	/api/v1/companies	GET	List all 92 companies with id, name, sector. Supports ?sector=
11.2	/api/v1/companies/{ticker}	GET	Full company profile: KPIs, pros/cons, sector, description.
11.3	/api/v1/companies/{ticker}/pl	GET	P&L history for a company. Supports ?from_year= ?to_year=
11.4	/api/v1/companies/{ticker}/bs	GET	Balance sheet history. Same year filters.
11.5	/api/v1/companies/{ticker}/cashflow	GET	Cash flow history. Same year filters.
11.6	/api/v1/companies/{ticker}/ratios	GET	All 14 pre-computed KPIs per year for one company.
11.7	/api/v1/companies/{ticker}/tearsheet	GET	Returns pre-generated tearsheet PDF (binary download).
11.8	/api/v1/screener	GET	Screener: ?min_roe=&max_de=§or=&min_cagr= → rank
11.9	/api/v1/sectors	GET	List all 11 sectors with company count and median KPIs.
11.10	/api/v1/sectors/{sector}/companies	GET	All companies in a sector with KPI summary.
11.11	/api/v1/peers/{group_name}	GET	All companies in a peer group with percentile ranks.
11.12	/api/v1/companies/{ticker}/peers/compare	GET	Radar data: company vs peer group average for 8 metrics.
11.13	/api/v1/market-cap/{ticker}	GET	Historical valuation multiples (P/E, P/B, EV/EBITDA) 2019–20
11.14	/api/v1/portfolio/stats	GET	Portfolio-level statistics: P10–P90 for all 10 core KPIs.
11.15	/api/v1/companies/{ticker}/documents	GET	Annual report links for a company.
11.16	/api/v1/health	GET	Server health check. Returns DB row counts and server upti

Output	Detail
api/main.py	FastAPI app. All routes in routers/ directory. One router file per module.
openapi.json	Auto-exported OpenAPI 3.0 spec. Import into Postman for API testing.

Module 12: Testing & Quality Assurance Module

pytest test suite with 60+ test cases covering: ETL correctness, KPI formula accuracy, API endpoint responses, data quality rule coverage. CI-style: all tests must pass before code is merged.

Sprint: Sprint 6 (Days 36–45)

#	Test Category	Test Cases	Pass Criterion
12.1	ETL Tests	year_normaliser (20 cases), ticker_normaliser (25 cases), test_p45 (5 cases), load_audit completeness (5 cases)	All 45 tests pass
12.2	KPI Formula Tests	ROE with positive equity, ROE with negative equity, None for interest-free (0), ICR for interest=0 (None),	100% accuracy
12.3	DQ Rule Tests	Each of 14 DQ rules triggered on crafted violation	14/14 rules trigger correctly
12.4	API Tests	/health returns 200; /companies returns 92 records; /sectors with invalid filters returns ranked list; invalid tick	All 10 API tests pass
12.5	Regression Tests	After any schema change, run full suite. No pre-existing test should fail.	Zero regressions

Output	Detail
tests/	pytest test directory. Structured: tests/etl/, tests/kpi/, tests/api/, tests/dq/
pytest_report.html	HTML report with pass/fail counts, runtime per test, failure traceback for failed tests

12. 45-Day Sprint Plan

6 Sprints × 7–8 Days
— Day-by-Day Task
Breakdown

Sprint	Days	Theme	Modules	Exit Criteria
Sprint 1	Days 01–07	Data Foundation	Module 1	nifty100.db built; all 10 tables loaded; load_audit.csv shows zero CRITICAL errors
Sprint 2	Days 08–14	Ratio Engine	Module 2	financial_ratios table populated for 92 × 14yr; 60+ KPIs computed; all 20 formula tests green
Sprint 3	Days 15–21	Screener & Peers	Modules 3–4	6 preset screeners operational; peer percentile table built for all 11 groups
Sprint 4	Days 22–28	Dashboard & Valuation	Modules 5–6	Streamlit app running on localhost:8501; all 8 screens navigable; valuation model deployed
Sprint 5	Days 29–35	Intelligence & Reports	Modules 7–9	92 company tearsheet PDFs generated; pros_cons_generated.csv complete
Sprint 6	Days 36–45	API, ML & QA	Modules 10–12	All 16 API endpoints return correct data; 60+ pytest tests pass; cluster_labels ML model trained

Sprint 1 — Data Foundation (Days 1–7)

Day	Tasks	Deliverable
D01	Set up project directory structure (data/, src/etl/, src/analytics/, src/api/, src/dashboard/). Create requirements.txt, install all dependencies; configure .env file.	Project structure ready; dependencies installed
D02	Write Excel loader with header=1 support for all 7 core files. Write normalize_year() and normalize_ticks() functions and tests: 20 cases each.	Excel loader v1.0; normalization functions
D03	Implement schema validator (16 DQ rules). Run against all 7 files. Log failures to validation_errors.csv. Review failures and lead.	Schema validator; validation_errors.csv
D04	Build SQLite schema (CREATE TABLE statements for 10 tables with FK constraints). Write loader scripts to load normalized DataFrames.	SQLite schema; loader scripts
D05	Load all 5 supplementary files. Run full load for all 12 files. Generate load_audit.csv.	nifty100.db, load_audit.csv
D06	Data quality review: manually check 5 random companies across all time-series tables. Document any issues in issues_log.md.	Manual review log
D07	Write 10 SQL exploratory queries (row counts, nulls, year coverage per company). Store queries in sql_explore/, task1_retro.md	Exploratory queries; sprint1_retro.md

Sprint 2 — Ratio Engine (Days 8–14)

Day	Tasks	Deliverable
D08	Implement profitability ratios: NPM, OPM, ROE, ROCE. Edge cases: negative equity (None), zero sales (None). Cross-validate OPM vs source.	Profitability ratios (None)
D09	Implement leverage + efficiency ratios: D/E (bank carve-out), ICR (debt-free substitution). Add 5 leverage/rpe tests.	Leverage/ICR tests
D10	Implement CAGR engine: Revenue CAGR, PAT CAGR, EPS CAGR for 3yr, 5yr, 10yr windows. Add 3 CAGR round flag logic.	CAGR engine
D11	Implement cash flow KPIs: FCF, CFO Quality Score, CapEx Intensity, FCF Conversion. Add 4 CF KPI tests (8 classes).	Cash flow KPIs (8 classes)
D12	Populate financial_ratios table in SQLite for all 92 companies × all years. Run validation on 10 random companies vs Excel calculations.	financial_ratios table
D13	Implement ROCE for banks/NBFCs: sector-relative approach. Cross-check vs roce_percentages_in_notebook.xlsx. Log anomalies.	ROCE for banks/NBFCs
D14	Run all 20 KPI formula tests. Fix failures. Write ratio_edge_cases.log. Sprint retrospective.	All KPI tests green; sprint2_retro.md

Sprint 3 — Screener & Peer Comparison (Days 15–21)

Day	Tasks	Deliverable
D15	Build custom filter engine: load screener_config.yaml; apply threshold filters to financial ratios via pandas query.	src/ratios/financial.py
D16	Implement 6 preset screeners. Test each preset on the full 92-company universe. Verify results against business logic.	src/screeners/presets.py
D17	Build ranking engine: composite score (50/30/20 weights). Sector-relative normalisation. Export top 10 per sector.	src/export/top_n.py, output_output.xlsx
D18	Build peer group module: load peer_groups.xlsx; compute PERCENT_RANK per metric. Apply to score for all peer groups.	src/peer_groups/peer_rank.py
D19	Generate radar chart data for each company-peer group pair. Plot Plotly radar PNG for each pair.	reports/radar_charts/ (92 PNGs)
D20	Generate peer_comparison.xlsx: one sheet per peer group, all metrics, colour-coded cells.	peer_comparison.xlsx
D21	Run all DQ rule tests. Fix failures. Sprint retrospective.	All tests green, sprint3_retro.md

Sprint 4 — Dashboard & Valuation (Days 22–28)

Day	Tasks	Deliverable
D22	Scaffold Streamlit app: main app.py + pages/ directory. Shared data loader (load_db_connection.py). Styling (sidebar logo, colour theme).	app.py, pages/, static/
D23	Build Home screen (5.1) and Company Profile screen (5.2): ticker search, KPI tiles, 10 charts.	pages/01_home.py, pages/02_profile.py
D24	Build Screener screen (5.3): sidebar sliders, live table, CSV download. Peer Comparison screen (5.4): group selector, radar.	pages/03_screener.py, pages/04_peer.py
D25	Build Trend Analysis (5.5), Sector Analysis (5.6), Capital Allocation Map (5.7), Annual Report (5.8)	pages/05_trends.py
D26	Build Valuation module: valuation_summary.xlsx, overvaluation flags, FCF yield column. Valuation ratio heatmap screen if time permits.	valuation_summary.xlsx, src/valuation.py
D27	Dashboard integration testing: test all 8 screens with 10 random tickers. Fix chart sizing, missing data edge cases.	Test log dashboard_integration.txt
D28	Sprint retrospective. Update README.md with app run instructions (streamlit run app.py). Demo and create valuation tool (optional).	README.md, valuation_tool.py (optional)

Sprints 5–6 — Intelligence, Reports, API & QA (Days 29–45) Summary

Days	Theme	Key Daily Tasks	Exit Gate
D29–30	NLP Module	Parser for analysis.xlsx CAGR text. Auto pros/cons generator (GPT-4o or LLaMA).	62 pros/cons generated for 92 companies
D31–32	Cash Flow Intelligence	CFO quality score, CapEx intensity, distress pattern flagging, capital allocation matrix.	cash_flow_intelligence.xlsx
D33–35	PDF Report Generator	Company Tearsheet (2-page × 92). Sector reports (11). Portfolio summary + All reports PDFs.	92 tearsheets, 11 sector reports, 1 portfolio summary, All reports PDFs/ dir.
D36–37	KMeans Clustering	StandardScaler + KMeans(n=5). Elbow plot. Assign cluster labels. Clustering heatmap.	cluster_labels.txt, heatmap.png
D38–40	FastAPI Server	16 endpoints in api/ directory. Uvicorn server. OpenAPI docs generated.	api/, uvicorn, openapi.json
D41–42	Full Test Suite	60+ pytest tests across 4 categories. Fix failures. Generate HTML report.	pytest report.html: all green
D43	Integration Testing	End-to-end: API → Dashboard data flow. Load test with 10 concurrent users.	Load test results: no errors, 6 bottlenecks.
D44	Documentation	Update all docstrings. Write analyst_guide.pdf (how to use the dashboard). Review project doc.	analyst_guide.pdf
D45	Final Sign-Off	Run acceptance checklist (23 gates). Present to team lead.	Signed acceptance checklist output/ dir.

13. KPI Reference — 50+ Financial Metrics

Formulas,
Benchmarks, Edge
Cases & Warning
Flags

KPI	Category	Formula	Benchmark	Edge Case / Warning Flag
Net Profit Margin	Profitability	$\text{net_profit} / \text{sales} \times 100$	>10% = good; >20% = excellent	Excludes sales=0. Negative allowed.
Operating Profit Margin	Profitability	$\text{operating_profit} / \text{sales} \times 100$	>15% = good; >25% = excellent	Check vs opm_percentage field ±1%.
EBIT Margin	Profitability	$(\text{operating_profit} - \text{depreciation}) / \text{sales} \times 100$	>12% preferred	Excludes other_income to get core ops only.
Return on Equity (ROE)	Returns	$\text{net_profit} / (\text{equity} + \text{reserves}) \times 100$	>15% = good; >20% = excellent	Excludes equity+reserves ≤ 0.
Return on Capital (ROCE)	Returns	$\text{EBIT} / (\text{equity} + \text{reserves} + \text{borrowings}) \times 100$	>15% = good; >25% = excellent	Exclude carve-out: use sector-relative benchmark.
Return on Assets (ROA)	Returns	$\text{net_profit} / \text{total_assets} \times 100$	>5% = good; >10% = excellent	Exclude if total_assets = 0.
Debt-to-Equity (D/E)	Leverage	$\text{borrowings} / (\text{equity} + \text{reserves})$	<1.0 = healthy; <0.5 = conservative	Flag >5 for non-financials.
Interest Coverage (ICR)	Leverage	$(\text{op_profit} + \text{other_income}) / \text{interest}$	>3× = safe; >5× = strong	None if interest=0 → display 'Debt Free'. Flag if < 3.
Net Debt	Leverage	$\text{borrowings} - \text{investments} - \text{cash}$	< 0 = net cash positive	investments used as liquid asset proxy.
Net Debt / EBITDA	Leverage	$\text{net_debt} / \text{operating_profit}$	<2× = healthy; >4× = stressed	Flag if EBITDA < 0.
Asset Turnover	Efficiency	$\text{sales} / \text{total_assets}$	>1× = efficient. Asset-light	None if total_assets = 0.
Fixed Asset Turnover	Efficiency	$\text{sales} / \text{fixed_assets}$	>3× = good	None if fixed_assets = 0 (service cos).
Working Capital Days	Efficiency	$(\text{other_asset} - \text{other_liabilities}) / \text{sales} \times 365$	Lower = better	>180 days = slow collections flag.
Inventory Turnover	Efficiency	$\text{sales} / (\text{fixed_assets proxy})$	Sector-dependent	Proxy only — no raw inventory field.
Revenue CAGR (3yr)	Growth	$(\text{sales}_t / \text{sales}_{\{t-3\}})^{(1/3)} - 1$	>10% = healthy; >20% = fast	Flag if base < 0.
Revenue CAGR (5yr)	Growth	$(\text{sales}_t / \text{sales}_{\{t-5\}})^{(1/5)} - 1$	>12% = healthy; >25% = strong	Flag if base < 0.
Revenue CAGR (10yr)	Growth	$(\text{sales}_t / \text{sales}_{\{t-10\}})^{(1/10)} - 1$	>10% sustained = excellent	Flag if <10yr history.
PAT CAGR (3yr)	Growth	$(\text{net_profit}_t / \text{net_profit}_{\{t-3\}})^{(1/3)} - 1$	>15% = healthy; >25% = strong	Flag if base < 0.
PAT CAGR (5yr)	Growth	$(\text{net_profit}_t / \text{net_profit}_{\{t-5\}})^{(1/5)} - 1$	>15% = healthy; >30% = excellent	Flag if base < 0.
EPS CAGR (5yr)	Growth	$(\text{eps}_t / \text{eps}_{\{t-5\}})^{(1/5)} - 1$	>12% = healthy	Must align with PAT CAGR unless share count ch.
Free Cash Flow	Cash Quality	$\text{operating_activity} + \text{investing_activity}$	> 0 = generating cash	Negative 3yr consecutive → 'FCF Concern' flag.
CFO / PAT Ratio	Cash Quality	$\text{operating_activity} / \text{net_profit}$	>1.0 = high quality earnings	Flag if < 0.5 = accrual risk. None if net_profit = 0.
CapEx Intensity	Cash Quality	$\text{abs}(\text{investing_activity}) / \text{sales} \times 100$	<3% = asset-light; >8% = capital intensive	Flag if > 10% — may include non-CapEx
FCF Conversion Rate	Cash Quality	$\text{FCF} / \text{operating_profit} \times 100$	>60% = efficient; <30% = poor	None if operating_profit = 0.
FCF Yield	Valuation	$\text{FCF} / \text{market_cap_crore} \times 100$	>3% = attractive	Requires market_cap table join.
Price-to-Earnings (P/E)	Valuation	$\text{market_cap} / \text{net_profit}$	15–25× = fair for India large caps	Flag if net_profit ≤ 0. Compare within sector.
Price-to-Book (P/B)	Valuation	$\text{market_cap} / (\text{equity} + \text{reserves})$	>3× = expensive for value	Varies widely by sector: IT >8×, utilities <2×.
EV/EBITDA	Valuation	$\text{enterprise_value} / \text{operating_profit}$	12–18× = fair for Indian large caps	Flag if EBITDA < 0.
Dividend Yield	Valuation	dividend_yield_pct from market_cap table	>2% = dividend-paying; >8% = high yield	Requires dividend table.
Dividend Payout Ratio	Profitability	dividend_payout from P&L table	30–60% = balanced	Flag >100% (paying more than earned).
Book Value Per Share	Valuation	$(\text{equity} + \text{reserves}) / (\text{equity_cap} / \text{face_value})$	>1 always. Rising trend	Computed — cross-check vs companies.book_val
Earnings Per Share (EPS)	Profitability	eps from P&L table	Rising trend preferred	Dilution risk if equity issuance ongoing.
Capital Allocation: Reinvestor	Qualitative	CFO>0, CFI<0, CFF<0	Ideal: paying debt + investing	State from ops –)
Capital Allocation: Shareholder Return	Qualitative	CFO>0, CFI<0, CFF<0 (dividends)	Stable mature company	Pattern: (+, –, –) same as above — sub-classify by
Capital Allocation: Distress	Qualitative	CFO<0, CFF>0	Raising funds to fund ops	Pattern (flag?, +)
Composite Quality Score	Composite	$0.3 \times \text{ROE_score} + 0.25 \times \text{FCF_score} + 0.25 \times \text{CFO_score} + 0.2 \times \text{P/B_score}$	0.0–1.0 scale	Normalized 0–100 using P10/P90 winsorisation.

14. Data Quality Rules

16 Validation Rules —
Applied at ETL Load
Time

Rule ID	Rule Name	Condition	Severity	Action
DQ-01	Company PK Uniqueness	len(companies) == companies.id.nunique()	CRITICAL	Halt load. Investigate duplicate ticker.
DQ-02	Annual PK Uniqueness	No duplicate (company_id, year) in P&L, BS, CF, OPM files	CRITICAL	Deduplicate: keep last occurrence. Log all duplicates.
DQ-03	FK Integrity	All company_id in child tables exist in companies	CRITICAL	Reject orphan rows. Log to validation_failures.csv.
DQ-04	Balance Sheet Balance	total_assets – total_liabilities / total_assets < 0.01	WARNING	Flag row. Do not reject. Analyst review required.
DQ-05	OPM Cross-Check	opm_percentage – (op_profit/sales×100) < 0.01	WARNING	Flag row. Use computed OPM in Ratio Engine; keep source for audit.
DQ-06	Positive Sales	sales > 0 for all non-bank companies	WARNING	Flag rows with sales ≤ 0. Exclude from growth CAGR.
DQ-07	Year Format	After normalize_year(), all values match r'^[0-9]{4}\$'	CRITICAL	Reject row if unparseable. Log raw value.
DQ-08	Ticker Format	company_id = company_id.strip().upper().length < 12	CRITICAL	Trim trailing spaces. If length out of range, reject.
DQ-09	Net Cash Check	net_cash_flow – (CFO+CFI+CFF) ≤ 10 (in ₹ Crores)	WARNING	Flag and compute net_cash_flow from components if mismatch.
DQ-10	Non-Negative Fixed Assets	fixed_assets ≥ 0	WARNING	Negative fixed_assets → coerce to 0 and log.
DQ-11	Tax Rate Range	0 ≤ tax_percentage ≤ 60	WARNING	Flag out-of-range. Can indicate one-off deferred tax reversal.
DQ-12	Dividend Payout Cap	dividend_payout ≤ 200 (pct)	WARNING	Flag >200% as likely data entry error. Analyst confirm.
DQ-13	URL Validity (documents)	requests.head(Annual_Report).status_code < 400	WARNING	Log 404 URLs. Do not reject row — URL decay expected over time.
DQ-14	EPS Sign Consistency	eps > 0 if net_profit > 0	WARNING	Flag mismatch (may indicate adjustments). Use net_profit/share for EPS.
DQ-15	BSE/ASE Balance (ext.)	total_liabilities == total_assets (strict, after DQ-04 flag)	INFO	Informational counter. Flag in load_audit only.
DQ-16	Coverage Check	Each company has ≥ 5 years of P&L, BS, CF, OPM	WARNING	Flag companies with < 5yr history. Exclude from CAGR if < 3yr.

15. API Specification

16 REST Endpoints — FastAPI / OpenAPI 3.0

Endpoint	Method	Query Params	Response Fields	Status Codes
/api/v1/companies	GET	sector, market_cap_category, search_string	search_string, name, broad_sector, sub_sector, company_id	200 OK, 400 Bad Request
/api/v1/companies/{ticker}	GET	year (optional; default=latest)	Full company object: all companies fields	200 OK, 404 Not Found
/api/v1/companies/{ticker}/pl	GET	from_year (YYYY-MM), to_year (YYYY-MM)	Array of rows for company. All 13 numerical fields	200 OK, 404 Not Found
/api/v1/companies/{ticker}/bs	GET	from_year, to_year	Array of balance sheet rows. All 11 numerical fields	200 OK, 404 Not Found
/api/v1/companies/{ticker}/cashflow	GET	from_year, to_year	Array of cash flow rows. CFO, CFI, CFF, 800 Crores	200 OK, 404 Not Found
/api/v1/companies/{ticker}/ratios	GET	year (optional; default=all)	Array: company_id, year + 14 computed ratios	200 OK, 404 Not Found
/api/v1/companies/{ticker}/tearsheet	GET	—	Binary PDF stream (application/pdf). Triggered on request	200 OK, 404 Not Found
/api/v1/screener	GET	min_roe, max_de, min_fcf, sector, min_revenue, company_id, ticker, 5yr, 10yr, 15yr, 20yr, 25yr, 30yr, 35yr, 40yr, 45yr, 50yr, 55yr, 60yr, 65yr, 70yr, 75yr, 80yr, 85yr, 90yr, 95yr, 100yr	Array of matching companies with composite score	200 OK, 400 Bad Request
/api/v1/sectors	GET	—	Array: sector_name, company_count, median_pe, median_de	200 OK
/api/v1/sectors/{sector}/companies	GET	year (default=latest)	Array of companies in sector with top 8 KPIs	200 OK, 404 Not Found
/api/v1/peers/{group_name}	GET	year (default=latest)	Array of companies + percentile rank for each KPI	200 OK, 404 Not Found
/api/v1/companies/{ticker}/peers/compare	GET	year (default=latest)	Radar data: 8 axis metrics for company + 200 peers	200 OK, 404 Not Found
/api/v1/market-cap/{ticker}	GET	from_year, to_year	Array: year, market_cap_crore, pe_ratio, dividend_yield	200 OK, 404 Not Found
/api/v1/portfolio/stats	GET	year (default=latest)	P10–P90 table for 10 KPIs across all 92 companies	200 OK
/api/v1/companies/{ticker}/documents	GET	from_year, to_year	Array: year, Annual_Report URL, is_url_valid	200 OK, 404 Not Found
/api/v1/health	GET	—	status, db_row_counts (all 10 tables), uptime	200 OK, 503 Service Unavailable

16. Risk Register

12 Project Risks —
Likelihood, Impact &
Mitigation Plan

Risk ID	Risk	Likelihood	Impact	Mitigation Strategy
R-01	Missing years of data for some companies (coverage < 5yr)	High	Medium	DQ-16 flags companies with < 5yr. Exclude from CAGR. Supplement with public filings
R-02	BSE Annual Report URLs returning 404 (link decay)	High	Low	DQ-13 validates URLs on load. Log failures. Dashboard shows 'Report unavailable' gra
R-03	analysis.xlsx text parsing fails for new formats	Medium	Medium	Regex tested for 5 format variants. Unknown formats logged to parse_failures.csv for m
R-04	Bank / NBFC D/E misclassification in screener	Medium	High	sectors.broad_sector used to carve out Financials from D/E filter. Tested on 19 financi
R-05	Negative equity companies breaking ROE formula	Medium	High	ROE = None if equity+reserves ≤ 0. Logged. Excluded from screener ROE filter.
R-06	Simulated stock_prices / market_cap values cause incorrect real conclusions	Low	High	All simulated datasets are clearly labelled SIMULATED in column comments and dash
R-07	CAGR turnaround flag not applied to misleading growth metrics	Low	High	CAGR engine explicitly checks base year sign. Turnaround flag stored in ratio column
R-08	ReportLab LayoutError on very long cells overflowing frames	Medium	Medium	All tables use WORDWRAP; cells > 200 chars are truncated with ellipsis. Tested on wo
R-09	Streamlit slows down with full 92-company dataset	Low	Medium	SQL queries are cached with @st.cache_data (TTL=600s). Max screener result set
R-10	Peer group coverage is partial (46/92 companies)	Medium	Medium	Peer module gracefully handles companies not in any group: returns 'No peer group as
R-11	pros/cons auto-generation produces low confidence	Medium	Low	Confidence score displayed alongside auto-generated entries. Threshold: only show if
R-12	FastAPI / Uvicorn port conflict on analyst machine	Low	Low	Default port 8000. .env file contains PORT variable. README documents port change

17. Deliverables Checklist

23 Mandatory
Deliverables —
Sign-Off Required on
Day 45

#	Deliverable	Format	Owner Sprint	Sign-Off Criteria
D-01	nifty100.db	SQLite file	S1	All 10 tables populated. rowcounts match source. FK constraints pass.
D-02	load_audit.csv	CSV	S1	All 12 files loaded. Zero CRITICAL failures.
D-03	validation_failures.csv	CSV	S1	All DQ rule violations documented with severity.
D-04	exploratory_queries.sql	SQL	S1	10+ queries covering coverage, nulls, year distribution.
D-05	financial_ratios table	SQLite table	S2	14 KPI columns. 1,184+ rows. KPI formula tests pass.
D-06	capital_allocation.csv	CSV	S2	All 92 companies × all years with pattern label.
D-07	screener_output.xlsx	Excel	S3	6 preset screener results with composite score + 20 KPIs.
D-08	screener_config.yaml	YAML	S3	All 15 filter thresholds defined. Analyst-editable.
D-09	peer_comparison.xlsx	Excel	S3	11 sheets. Colour-coded percentile cells. 20 metrics.
D-10	radar_charts/ (92 PNGs)	PNG files	S3	One radar chart per company. 8 axes. Peer avg overlay.
D-11	Streamlit Dashboard (app.py)	Python app	S4	8 screens. All navigable. CSV export functional.
D-12	valuation_summary.xlsx	Excel	S4	All 92 companies. P/E, P/B, EV/EBITDA, flags.
D-13	cashflow_intelligence.xlsx	Excel	S5	CFO quality score, FCF CAGR, CapEx intensity, distress flags.
D-14	pros_cons_generated.csv	CSV	S5	92 companies. Rule-triggered. Confidence > 60%.
D-15	analysis_parsed.csv	CSV	S5	Structured CAGR numbers from analysis.xlsx. Cross-validated.
D-16	Company Tearsheets (92 PDFs)	PDF files	S5	2 pages per company. All 92 companies. In reports/tearsheets/
D-17	Sector Reports (11 PDFs)	PDF files	S5	One per broad sector. KPI table + company list.
D-18	Portfolio Summary PDF	PDF	S5	All 92 companies, 1-page each in one document.
D-19	cluster_labels.csv	CSV	S6	5 clusters. All 92 companies assigned. Cluster names labelled.
D-20	FastAPI Server (api/)	Python app	S6	16 endpoints. openapi.json exported. /health returns 200.
D-21	pytest_report.html	HTML	S6	60+ tests. All green. Runtime < 120 seconds total.
D-22	analyst_guide.pdf	PDF	S6	Screener usage guide + dashboard navigation. 10–15 pages.
D-23	acceptance_checklist.pdf	PDF	S6	23 deliverables reviewed and signed off. Date stamped.

18. Acceptance Criteria

20 Non-Negotiable
Quality Gates — All
Must Pass Before
Sign-Off

Gate	Area	Criterion	Tested By
AC-01	Data Coverage	92 companies present in companies table. No extra or missing tickers	SQL COUNT(*) FROM companies → 92
AC-02	Time Coverage	≥ 90% of companies have ≥ 10 years of P&L, BS, CF records	DQ coverage query in load_audit.csv
AC-03	Schema Integrity	All FK relationships intact. PRAGMA foreign_key_check returns 0 rows	SQL pragma check command
AC-04	KPI Completeness	financial_ratios table has ≥ 1,100 rows. All 14 KPI columns populated	SQL COUNT(*) FROM financial_ratios
AC-05	CAGR Accuracy	Revenue CAGR for 3 manually-checked companies matches Manual Exported Excel value ±0.1%.	Manual Exported Excel
AC-06	ROE Accuracy	ROE for 5 companies in latest year matches companies.roeM (percentage ±5% allow for version diff).	Manual Exported Excel
AC-07	Screener Accuracy	Quality preset screener (ROE>15, D/E<1, FCF>0) produces Screener results 500 companies.	Manual Exported Excel
AC-08	Dashboard Load	Company Profile screen loads for any of 92 tickers in < 3 seconds	Manual stopwatch test
AC-09	Dashboard Export	CSV download on Screener screen produces well-formed file	Downloaded and opened
AC-10	PDF Quality	No text overflow, no overlapping pages in any generated reports	Visual review of 5 random tearsheets
AC-11	API Health	GET /api/v1/health returns HTTP 200 with db_row_counts for all APIs	API health
AC-12	API Accuracy	GET /api/v1/companies/TCS/ratios returns rows for ≥ 10 years	API response check
AC-13	API Screener	GET /api/v1/screener?min_roe=15&max_de=1 returns consistent results with DQ output.	Response vs DQ output
AC-14	Peer Coverage	Peer percentile table populated for all 11 peer groups. All metrics populated	SQL COUNT(*) FROM peer_group_name → 11
AC-15	Cluster Coverage	All 92 companies assigned to a cluster (0–4). No nulls in cluster	tested tables: csv null check
AC-16	NLP Coverage	pros_cons_generated.csv has ≥ 1 pro and ≥ 1 con for every company	CSV parity check per company_id
AC-17	Report Coverage	92 tearsheet PDFs exist in reports/tearsheets/. Each is ≥ 50KB	ls reports/tearsheets/ wc -l → 92
AC-18	Test Coverage	pytest count shows ≥ 60 tests collected. 0 failures. 0 errors. pytest --tb=short output	
AC-19	DQ Documentation	validation_failures.csv exists. All rows have company_id, field, issue, severity	File presence, column check
AC-20	Documentation	analyst_guide.pdf exists and is ≥ 10 pages. Contains screener	File size, last page section check

19. Project Directory Structure

Standard Layout — All
Code, Data, Reports &
Tests

Path	Type	Contents
data/raw/	Directory	7 core Excel files (READ ONLY). Never modify.
data/supporting/	Directory	5 supplementary Excel files.
data/nifty100.db	SQLite	Primary database. All 10 tables.
src/etl/	Directory	loader.py, validator.py, normaliser.py, schema.sql
src/analytics/	Directory	ratios.py, cagr.py, cashflow_kpis.py, screener/engine.py, peer.py, clustering.py
src/nlp/	Directory	parser.py (analysis.xlsx), pros_cons_generator.py
src/dashboard/	Directory	app.py, pages/ (01–08), utils/charts.py, utils/db.py
src/api/	Directory	main.py, routers/ (companies, screener, peers, sectors, reports, health)
src/reports/	Directory	tearsheet.py, sector_report.py, portfolio_report.py, screener_report.py
tests/	Directory	etl/, kpi/, api/, dq/ — 60+ pytest test files
config/	Directory	screener_config.yaml, .env.template, logging_config.yaml
reports/tearsheets/	Directory	92 company tearsheet PDFs
reports/sector/	Directory	11 sector PDFs
reports/portfolio/	Directory	Portfolio summary PDFs (weekly)
reports/radar_charts/	Directory	92 radar chart PNGs
output/	Directory	Screener exports, ad hoc CSVs, final deliverables archive
notebooks/	Directory	Exploratory Jupyter notebooks (not in production path)
docs/	Directory	Project document PDF, analyst guide, openapi.json
requirements.txt	File	All pip dependencies pinned to minor version
.env	File	DB_PATH, PORT, LOG_LEVEL, SIMULATED_DATA_FLAG. Not committed to git.
README.md	File	Project overview, setup instructions, how to run app and API
Makefile	File	make load, make test, make report, make api, make dashboard

20. Glossary

Financial & Technical
Terms Used
Throughout This
Document

Term	Definition
Asset Turnover	Sales divided by total assets. Measures how efficiently a company uses its assets to generate revenue.
Book Value Per Share	(Equity capital + Reserves) divided by number of shares outstanding. Represents net asset value per share.
CAGR	Compound Annual Growth Rate. $((\text{End Value} / \text{Start Value})^{(1/n)} - 1) \times 100$. Annualised growth metric.
CapEx	Capital Expenditure. Funds used to acquire or upgrade long-term physical assets. Proxy: absolute value of investing_activity.
Capital Allocation	How a company distributes its generated cash between reinvestment, debt repayment, and shareholder returns.
CFI	Cash Flow from Investing Activities. Covers CapEx, acquisitions, and investment purchases. Typically negative.
CFF	Cash Flow from Financing Activities. Covers debt issuance/repayment and equity transactions including dividends.
CFO	Cash Flow from Operations. Cash generated by core business activities. Must be positive for a healthy business.
CWIP	Capital Work-In-Progress. Assets under construction not yet in use. Excluded from net fixed assets.
D/E	Debt-to-Equity Ratio. Total borrowings divided by total equity (capital + reserves). Measures leverage.
Debt-Free	Company with zero or negligible borrowings. D/E = 0. ICR displayed as 'Debt Free' instead of infinity.
EBIT	Earnings Before Interest and Taxes. Operating_profit minus depreciation. Core operational earnings metric.
EBITDA	Earnings Before Interest, Taxes, Depreciation & Amortisation. Proxy: operating_profit in this dataset.
EPS	Earnings Per Share. Net profit divided by number of shares. Source: eps column in profitandloss table.
EV	Enterprise Value. Market Cap + Net Debt. Represents total company value to all capital providers.
EV/EBITDA	Enterprise Value divided by EBITDA. Cross-sector valuation multiple. Lower = potentially cheaper.
FCF	Free Cash Flow. CFO + CFI. Cash available after maintaining/expanding asset base. Core quality metric.
FCF Yield	FCF divided by Market Cap $\times 100$. Higher = better value for cash-generating companies.
FK	Foreign Key. A database constraint linking child table rows to parent table rows via a shared key column.
FY	Financial Year. In India, April 1 to March 31. FY24 = April 2023 to March 2024. Labelled 'Mar-24' in source data.
GICS	Global Industry Classification Standard. Tiered sector taxonomy: Sector $\rightarrow$ Industry Group $\rightarrow$ Industry $\rightarrow$ Sub-Industry.
ICR	Interest Coverage Ratio. $(\text{EBIT} + \text{Other Income}) / \text{Interest expense}$. Measures ability to service debt.
KPI	Key Performance Indicator. A quantifiable measure used to evaluate business performance. 50+ defined in Section 13.
Large Cap	Companies with market capitalisation $> \text{₹}20,000$ Crore. All Nifty 100 companies qualify.
Nifty 100	NSE India index of 100 largest companies by free-float market cap. This dataset covers 92 of those companies.
NPM	Net Profit Margin. Net profit divided by total sales $\times 100$. Expressed as percentage.
NBFC	Non-Banking Financial Company. Lender/financial services firm that is NOT a bank. High D/E structurally normal.
NSE	National Stock Exchange of India. Primary stock exchange. Ticker symbols used as company primary keys.
OPM	Operating Profit Margin. Operating profit divided by sales $\times 100$. EBITDA margin proxy.
P/B	Price-to-Book Ratio. Market Cap divided by Book Value (equity + reserves). >1 means market premium to assets.
P/E	Price-to-Earnings Ratio. Market Cap divided by Net Profit. Standard equity valuation multiple.
PAT	Profit After Tax. Same as Net Profit in this dataset.
PBT	Profit Before Tax. Net Profit divided by $(1 - \text{tax_percentage}/100)$. Pre-tax earnings.
ROCE	Return on Capital Employed. $\text{EBIT} / (\text{Equity} + \text{Borrowings}) \times 100$. Measures returns on all capital, not just equity.
ROE	Return on Equity. Net Profit $/ (\text{Equity} + \text{Reserves}) \times 100$. Measures returns on shareholders' funds.
SQLite	Serverless, file-based relational database. Single .db file. Zero configuration. Used for this project's data store.

Term	Definition
Turnaround Flag	Applied to CAGR calculations where the base year value is negative and end year value is positive. CAGR set to None.
UW	Usable Width. PDF page width (21 cm) minus margins (3 cm) = 16.6 cm. All table column widths must sum to UW.
Winsorisation	Statistical technique: cap extreme values at P10/P90. Prevents outliers from distorting composite score normalisation.

21. Environment Setup & Quick-Start Guide

From Clone to
Running App in Under
30 Minutes

Step	Command / Action	Expected Output
1. Clone / unzip project	unzip nifty100_project.zip -d nifty100/	Project directory created with data/, src/, tests/, config/
2. Create virtual environment	python3 -m venv .venv && source .venv/bin/activate	(venv) prompt in terminal
3. Install dependencies	pip install -r requirements.txt	All 20 libraries installed; no errors
4. Configure environment	cp config/.env.template .env && nano .env	Set DB_PATH=data/nifty100.db; PORT=8000; LOG_LEVEL=INFO
5. Build database (ETL)	python src/etl/loader.py (or: make load)	nifty100.db created; load_audit.csv shows 12 files loaded
6. Run KPI computation	python src/analytics/ratios.py (or: make ratios)	financial_ratios table populated; 1,100+ rows
7. Run test suite	pytest tests/ --html=reports/pytest_report.html (or: make test)	60+ tests passed; 0 failures
8. Generate PDF reports	python src/reports/portfolio_report.py (or: make report)	reports/tearsheets/ populated with 92 PDFs
9. Start Streamlit dashboard	streamlit run src/dashboard/app.py (or: make dashboard)	Dashboard opens at http://localhost:8501
10. Start REST API	uvicorn src.api.main:app --port 8000 (or: make api)	API running at http://localhost:8000/docs

Makefile Target	What it does	When to use
make load	Runs ETL: loads all 12 files into nifty100.db. Re-runnable (idempotent).	Day 1 and after any source file update
make ratios	Runs Ratio Engine: populates financial_ratios table.	After make load. Re-run after any KPI formula change.
make test	Runs full pytest suite. Generates HTML report.	Before every commit. Must be 0 failures.
make report	Generates all PDF reports: 92 tearsheets + 11 sector + 1 portfolio.	Sprint 5+. Run once a week for fresh reports.
make dashboard	Starts Streamlit on port 8501.	Day-to-day analytics work.
make api	Starts FastAPI/Uvicorn on port 8000.	Sprint 6+. When dashboard needs live API.
make clean	Deletes .pyc files, __pycache__, and test artifacts (NOT the database).	Before packaging final deliverables.

22. Coding Standards & Best Practices

Mandatory Conventions for All Submitted Code

Area	Standard	Tooling / Reference
Formatting	Black — line length 88. No manual formatting.	black src/ tests/ — run pre-commit
Linting	ruff — all default rules enabled. Zero linting errors before commit.	ruff check src/ tests/
Type Hints	All function signatures must have type annotations. Return types mandatory (optional in S1; mandatory S4+)	mypy (optional)
Docstrings	One-line docstring for all public functions. No multi-paragraph docstrings.	Example: def load_pl(path: str) -> pd.DataFrame: """Load profitandloss E
Imports	stdlib → third-party → local. One blank line between groups. ruff enforced.	from pandas import DataFrame
Naming	snake_case for functions and variables. PascalCase for classes. UPPER_CASE for module-level constants.	PERC_CAGR for module-level constants.
Configuration	All thresholds in screener_config.yaml or .env. Zero hardcoded values in the analytics or app YAML	python analytics/ or app/
Error Handling	Wrap external I/O (file reads, URL fetches) in try/except. Log error; don't raise unless user sees logging_config.yaml	logging module
Logging	Use Python logging (not print). Levels: DEBUG for ETL detail; INFO for product logging; WARN for config issues; ERROR for failures.	logging module
Database	Use context manager (with sqlite3.connect(DB) as conn) for all queries. Parameterize (SELECT FROM + WHERE) SQL. (ticker,))	Parameterize (SELECT FROM + WHERE) SQL. (ticker,))
Testing	One test file per source module. Test file: tests/etl/test_loader.py mirrors code locations.py. fixtures in confest.py	pytest module
Git	Commit message format: [S1] feat: load P&L from Excel — sprint tag. No type as in commit. Max 72 chars.	Notype as in commit. Max 72 chars.

23. ETL Edge Cases — Complete Reference

Source Field	Raw Value	normalize_year() Output	Action
year	Mar-23	2023-03	Standard format — most common
year	Mar 23	2023-03	Strip space before parse
year	March-2023	2023-03	Full month name support
year	2023	2023-03	Integer year → assume March FY close
year	FY23	2023-03	FY prefix removal
year	Dec-22	2022-12	December year-end company (e.g. NESTLEIND)
year	Jun-23	2023-06	June year-end (some banks)
year	2023-03	2023-03	Already normalised — pass through
year	garbage	PARSE_ERROR	Reject row; log raw value to parse_failures.csv
company_id	TCS	TCS	Strip whitespace both sides
company_id	tcs	TCS	Upper-case
company_id	BAJAJ-AUTO	BAJAJ-AUTO	Hyphen preserved — valid NSE ticker
company_id	M&M	M&M	Ampersand preserved — valid NSE ticker
company_id		MISSING	Reject row — no FK match possible

23.1 CAGR Edge Case Decision Table

Base Year Value	End Year Value	CAGR Result	Flag	Display As
Positive (> 0)	Positive (> 0)	Computed normally	—	e.g. 18.3%
Positive (> 0)	Negative (< 0)	None	DECLINE_TO_LOSS	N/A — turned loss
Negative (< 0)	Positive (> 0)	None	TURNAROUND	Turnaround ↑
Negative (< 0)	Negative (< 0)	None	BOTH_NEGATIVE	N/A — both loss
Zero	Any	None	ZERO_BASE	N/A — base=0

Base Year Value	End Year Value	CAGR Result	Flag	Display As
< 3yr history	Any	None	INSUFFICIENT	N/A — < 3yr

24. Sample SQL Queries

Ready-to-Run Queries
for Common Analytical
Tasks

Query Name	SQL	Purpose
Top 10 ROE Companies (Latest Year)	SELECT r.company_id, c.company_name, r.return_on_equity_pct FROM financial_ratios r JOIN companies c ON r.company_id=c.id WHERE r.year=2024 ORDER BY r.return_on_equity_pct DESC	Identify highest ROE companies for quality analysis
Debt-Free Companies	SELECT company_id, year, debt_to_equity FROM financial_ratios WHERE debt_to_equity=0 AND year=(SELECT MAX(year) FROM financial_ratios)	Filter zero-debt companies (Safe ROE/ACX/year)
Consecutive FCF Positive (5yr)	SELECT company_id, COUNT(*) AS positive_fcf_yrs FROM financial_ratios WHERE free_cash_flow > 0 GROUP BY company_id HAVING COUNT(*) >= 5	Identify consistently profitable companies
Sector Median ROE	SELECT s.broad_sector, ROUND(AVG(r.return_on_equity_pct),1) AS median_roe FROM financial_ratios r JOIN sectors s ON r.company_id=s.broad_sector GROUP BY s.broad_sector	Sum of ROE from financial ratios per sector
Capital Allocation Pattern Count	SELECT pattern_label, COUNT(*) AS companies FROM capital_allocation WHERE year=(SELECT MAX(year) FROM capital_allocation) GROUP BY pattern_label	Count by year of Capital Allocation Patterns (Nifty 100)
Revenue CAGR > 15% (5yr)	SELECT r.company_id, c.company_name, r.revenue_cagr_5yr FROM financial_ratios r JOIN companies c ON r.company_id=c.id WHERE r.revenue_cagr_5yr > 0.15	Highlight high-growth companies
Missing Annual Reports	SELECT c.id, c.company_name, 2024-COUNT(d.Year) AS missing_yrs FROM companies c LEFT JOIN documents d ON c.id=d.company_id WHERE d.Year IS NULL	Identify companies with missing documents
Peer Group Rankings (ROE)	SELECT p.peer_group_name, r.company_id, r.return_on_equity_pct, RANK() OVER (PARTITION BY peer_group_name ORDER BY r.return_on_equity_pct DESC) AS rank FROM peer_groups p JOIN financial_ratios r ON p.company_id=r.company_id	Over-ROE Ranking by peer groups

25. Screener Configuration Reference

screener_config.yaml
— All Threshold Definitions

Preset Name	Filters Applied	Ranking Metric	Expected Company Count
Quality Compounder	ROE > 15%, D/E < 1.0, FCF > 0, Revenue CAGR 5yr > 10%	Composite Score (desc)	15–35 companies
Value Pick	P/E < 20, P/B < 3.0, D/E < 2.0, Dividend Yield > 1%	FCF Yield (desc)	10–25 companies
Growth Accelerator	PAT CAGR 5yr > 20%, Revenue CAGR 5yr > 15%, D/E < 2.0	PAT CAGR 5yr (desc)	8–20 companies
Dividend Champion	Dividend Yield > 2%, Dividend Payout < 80%, FCF > 0	Dividend Yield (desc)	10–20 companies
Debt-Free Blue Chip	D/E = 0, ROE > 12%, Revenue > 5,000 Cr	ROE (desc)	15–30 companies
Turnaround Watch	Revenue CAGR 3yr > 10%, FCF improving (positive latest year), Debt/Equity < 1.0	Revenue CAGR 3yr (desc)	5–15 companies

25.1 Composite Score Weights

Dimension	Weight	Inputs	Normalisation
Profitability	35%	ROE (15%), ROCE (10%), NPM (10%)	Winsorise P10–P90; scale 0–100 per metric; weighted avg
Cash Quality	30%	FCF CAGR 5yr (15%), CFO/PAT ratio (10%), FCF Margin (5%)	Same flag (5%) flag = 100 if positive, 0 if negative
Growth	20%	Revenue CAGR 5yr (10%), PAT CAGR 5yr (10%)	Same; CAGR flags (Turnaround) → score = 0
Leverage	15%	D/E score (10%), ICR score (5%)	D/E: 0=100, 0.5=85, 1=70, 2=50, >5=0. ICR: >10=100, 5=75, 3=50, <1.5=0

26. Dashboard Screen Specifications

Screen	URL Path	Key Data Queries	Interactive Elements	Export
Home / Overview	/	SELECT from companies + financial_ratios (last 5 years) + sector_classification (2019–2024)	Search box, year range slider	—
Company Profile	/company	All tables for selected ticker. Union P&L + Balance Sheet	Ticker search box, year range slider	PDF tearsheet download link
Financial Screener	/screener	SELECT from financial_ratios + market_cap_metrics WHERE (filters)	Filters sidebar, preset dropdowns	CSV download, Excel download
Peer Comparison	/peers	SELECT from peer_percentiles + financial_ratios (last 5 years)	Peer group dropdown, metric axis selector	PDF download, Excel table
Trend Analysis	/trends	SELECT time-series from financial_ratios (last 10 years)	Ticker search, metric multi-select (overlay)	CSV download
Sector Analysis	/sectors	SELECT with JOIN sectors. Bubble chart	Sector dropdown, year slider, metric selector	PDF download, Excel save
Capital Allocation Map	/capital	SELECT capital_allocation by year.	Year slider, click on pattern category → company list	CSV download
Annual Reports	/documents	SELECT from documents WHERE company_id = [selected]	Keyword search, year filter	—

27. Testing Framework — Test Case Reference

Test File	Test Name	Input	Expected Output
tests/etl/test_normalise.py	test_year_mar23	Mar-23	2023-03
tests/etl/test_normalise.py	test_year_fy24	FY24	2024-03
tests/etl/test_normalise.py	test_year_dec22	Dec-22	2022-12
tests/etl/test_normalise.py	test_year_garbage	xyz	PARSE_ERROR
tests/etl/test_normalise.py	test_ticker_strip	TCS	TCS
tests/etl/test_normalise.py	test_ticker_lower	tcs	TCS
tests/kpi/test_ratios.py	test_roe_positive	net_profit=100, equity=500	ROE = 20.0
tests/kpi/test_ratios.py	test_roe_neg_equity	net_profit=100, equity=-50	ROE = None
tests/kpi/test_ratios.py	test_de_debtfree	borrowings=0, equity=500	D/E = 0
tests/kpi/test_ratios.py	test_icr_debtfree	interest=0	ICR = None (display 'Debt Free')
tests/kpi/test_ratios.py	test_cagr_turnaround	base=-100, end=200	CAGR = None, flag='TURNAROUND'

Test File	Test Name	Input	Expected Output
tests/kpi/test_ratios.py	test_cagr_normal	base=100, end=161, n=5	CAGR ≈ 10.0%
tests/dq/test_rules.py	test_dq04_bs_balance	assets=1000, liab=1020	DQ-04 WARNING triggered
tests/dq/test_rules.py	test_dq06_zero_sales	sales=0	DQ-06 WARNING triggered
tests/api/test_health.py	test_health_200	GET /api/v1/health	HTTP 200, status='ok'
tests/api/test_api.py	test_companies_count	GET /api/v1/companies	92 records returned
tests/api/test_api.py	test_invalid_ticker	GET /api/v1/companies/INVALID	HTTP 404
tests/api/test_api.py	test_screener_filter	GET /api/v1/screener?min_roe=15	25 results have ROE ≥ 15

28. Sector Benchmarks & Expected Metric Ranges

Context for Anomaly Detection & Relative Scoring

Sector	ROE Range	D/E Range	OPM Range	P/E Range	Notes
Financials (Banks)	12–22%	> 5× (normal)	25–45% NIM proxy	12–25×	D/E structurally high — use NIM + ROA for banks instead
Financials (NBFC)	12–20%	3–8× (normal)	30–55% NIM proxy	15–30×	Higher ROA and ROE expected for consumer finance vs banks
Information Technology	25–50%	0–0.3×	20–35%	22–40×	Asset-light; high ROE; low D/E; premium P/E for growth
Consumer Staples (FMCG)	20–45%	0–0.5×	15–28%	40–70×	Very high P/E; brand moat; consistent margins; low debt
Healthcare / Pharma	15–30%	0–0.8×	15–25%	25–45×	Moderate debt; R&D adds intangible assets; premium for pipeline
Energy (Oil & Gas)	10–18%	0.5–2×	8–18%	8–15×	Capital intensive; commodity-linked margins; PSU major players
Energy (Power)	8–15%	1–4×	25–40%	12–20×	High D/E normal for regulated utilities; stable margin
Consumer Discretionary	10–25%	0.3–1.5×	10–20%	25–50×	Wide range; auto OEMs have lower OPM than consumer goods
Materials (Steel)	10–25%	0.3–1×	12–22%	8–18×	Cyclical; margins volatile with commodity prices
Materials (Cement)	12–22%	0.3–1×	18–28%	18–35×	Oligopoly; regional pricing power; moderate debt
Industrials	12–22%	0.3–1.5×	10–20%	20–35×	Capital goods order-book driven; L&T sets benchmark
Conglomerates	10–20%	0.5–2×	8–18%	15–30×	Diverse revenue streams; holding company discount applied

Usage: For relative scoring, normalise KPIs against these sector ranges rather than the universal Nifty 100 range. A D/E of 4× is a CRITICAL flag for an IT company but completely normal for a bank. The Ratio Engine uses sectors.broad_sector to select the appropriate benchmark range for each metric.

29. Week-by-Week Communication & Review Plan

Milestone Reviews, Deliverable Handoffs & Sign-Off Schedule

Week	Days	Review Meeting	Deliverables to Share	Go / No-Go Criteria
Week 1	D01–07	End of D07 — Sprint 1 Review (30 min)	nifty100.db, load_audit.csv, validation_failures.log, deployment_log.txt	All 100 cases deployed, zero CRITICAL DQ failures. Coverage > 90%
Week 2	D08–14	End of D14 — Sprint 2 Review (45 min)	financial_ratios table (SQLite export), ratio_engine_cases.log, initial_test_results.csv	100+ cases ingested, all tests pass. Formula spot-checks passed
Week 3	D15–21	End of D21 — Sprint 3 Review (45 min)	screener_output.xlsx (6 presets), peer_comparison_output.json, cluster_labels.log	11 peer companies added. All DQ checks pass
Week 4	D22–28	End of D28 — Sprint 4 Demo (60 min)	live Streamlit dashboard demo, valuation_report.pdf, error_logs.log	All 80 records load without errors. Screener returns correct results
Week 5	D29–35	End of D35 — Sprint 5 Review (45 min)	sample_tearsheets (3 companies), pros_cons_tearsheets.docx, model_performance_metrics.csv	Intelligence coverage > 90%. Model accuracy > 85%
Week 6	D36–42	End of D42 — Sprint 6 Review (60 min)	API demo (Postman), pytest_report.html, cluster_labels.csv	60+ tests pass. Cluster labels make business sense
Final	D43–45	D45 — Project Sign-Off (90 min)	All 23 deliverables, acceptance_checklist.pdf	All 30 acceptance gates pass. Lead signs acceptance checklist

29.1 Daily Stand-Up Format (5 minutes per person)

Question	Guidance	Max Time
What did I complete yesterday?	Name the specific deliverable or sub-task completed. Reference the Streamlit dashboard item.	90 seconds
What will I complete today?	Commit to a specific deliverable — not vague ('work on module X'). Name the file/function.	90 seconds
What is blocking me?	Only blockers that cannot self-resolve within 2 hours. Escalate to team lead if unresolved.	60 seconds
Anything to share?	Optional: share a finding, risk spotted, or cross-team dependency discovered.	60 seconds

Appendix A — Dataset File Checksums & Version Control

File Integrity
Reference for Source
Data

File	Size (approx)	Rows	SHA-256 (run at project start)	Notes
companies.xlsx	~50 KB	92	Run: sha256sum data/raw/companies.xlsx	Core reference — never modify
profitandloss.xlsx	~180 KB	1,276	Run: sha256sum data/raw/profitandloss.xlsx	Core time-series — never modify
balancesheet.xlsx	~190 KB	1,312	Run: sha256sum data/raw/balancesheet.xlsx	Core time-series — never modify
cashflow.xlsx	~130 KB	1,187	Run: sha256sum data/raw/cashflow.xlsx	Core time-series — never modify
analysis.xlsx	~20 KB	20	Run: sha256sum data/raw/analysis.xlsx	Partial coverage — never modify
documents.xlsx	~200 KB	1,585	Run: sha256sum data/raw/documents.xlsx	URL repository — never modify
prosandcons.xlsx	~15 KB	16	Run: sha256sum data/raw/prosandcons.xlsx	Partial coverage — never modify
sectors.xlsx	~20 KB	92	Run: sha256sum data/supporting/sectors.xlsx	Created — update if sector changes
stock_prices.xlsx	~400 KB	5,520	Run: sha256sum data/supporting/stock_prices.xlsx	Simulated — label in dashboard
market_cap.xlsx	~60 KB	552	Run: sha256sum data/supporting/market_cap.xlsx	Simulated — label in dashboard
financial_ratios.xlsx	~160 KB	1,184	Run: sha256sum data/supporting/financial_ratios.xlsx	Computed — regenerate from DB
peer_groups.xlsx	~15 KB	56	Run: sha256sum data/supporting/peer_groups.xlsx	Created — update if groups change

Appendix B — Contact & Escalation Matrix

Role	Responsibility	Response SLA	Escalation Path
Team Lead / Project Manager	Sprint planning, risk decisions, final sign-off on deliverables	Same day	—
Data Engineer	ETL module (Module 1), database schema, DQ rules, pipeline orchestration	4 hrs, pipeline criticality	Escalate to Team Lead after 2hr block
Analytics Engineer	Ratio Engine (Module 2), Screener (Module 3), Score Engine (Module 4)	Same Sprint day	Escalate to Team Lead for business logic decisions
Dashboard Developer	Streamlit app (Module 5), chart design, UX	Same sprint day	Escalate for data availability issues → Data Engineer
API Developer	FastAPI server (Module 11), OpenAPI docs, Postman collection	Same sprint day	Escalate for DB access issues → Data Engineer
QA / Testing	Test suite (Module 12), acceptance criteria verification, sign-off checklist	2 hrs prior to release	Escalate to Team Lead for acceptance gate failures

Document Sign-Off

Role	Name	Signature	Date
Project Manager / Team Lead		_____	_____
Data Engineering Lead		_____	_____
Analytics Lead		_____	_____
QA Lead		_____	_____

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